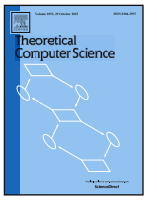
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Identity-based signature and identity-based matchmaking encryption from lattices

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ABSTRACT

We construct a lattice-based identity-based signature (IBS) scheme that achieves existential unforgeability under chosen-identity and chosen-message attacks (EUF-ID-CMA) in the standard model. Note that existing IBS schemes from lattices are either based on the random oracle model, only achieve selective security in the standard model, or only have a variant EUF-ID-CMA (denoted $\overline{\text{EUF-ID-CMA}}$) security. The $\overline{\text{EUF-ID-CMA}}$ security model requires the user to obtain a *fresh* signing key for signing each message, and thus it can hardly capture real-world applications. Therefore, our IBS scheme serves as the first EUF-ID-CMA secure IBS from lattices in the standard model.

We also propose a new framework of constructing identity-based matchmaking encryption (IB-ME) from IBS and identity-based encryption (IBE). We modify the stronger authenticity for IB-ME (Wang et al. TCS 2025) to a new stronger authenticity. We prove that the new stronger authenticity of IB-ME can be reduced to the EUF-ID-CMA security of the underlying IBS and the ciphertext pseudo-randomness of IB-ME can be reduced to the ciphertext pseudo-randomness of the underlying IBE. Our lattice-based IBS scheme and the IBE scheme (Boyen and Li AsiaCrypt 2016) yield the first lattice-based IB-ME scheme with new stronger authenticity in the standard model.

1. Introduction

Identity-based cryptography was first proposed by Shamir in 1984 [1], who introduced both identity-based encryption (IBE) and identity-based signature (IBS). Later, Ateniese et al. [2] proposed a new primitive named identity-based matchmaking encryption (IB-ME). The core idea of identity-based cryptography is to use a user's public identity information (for example, an email address, national ID number, or phone number) directly as the public key, while a trusted key generation center (KGC) computes and distributes the corresponding private key for that identity over a secure channel. Unlike traditional public-key infrastructure (PKI), identity-based schemes eliminate the need for digital certificates, thereby significantly simplifying key management. Consequently, identity-based cryptography has found wide application in scenarios such as encrypted communication and digital signatures.

Identity-Based Signatures. In an IBS system, the private key of each user is issued by a KGC and the user uses the private key to produce publicly verifiable signatures. The standard security notion for IBS is existential unforgeability under chosen-identity and chosen-message attacks (EUF-ID-CMA) [3,4]. In this model, the adversary is allowed to adaptively query the key generation oracle

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$\mathcal{O}_{\mathsf{KeyGen}}$ and the signing oracle $\mathcal{O}_{\mathsf{Sign}}$. EUF-ID-CMA security requires that no adversary can forge a valid signature on a new message without the corresponding secret signing key.

There are two generic approaches to construct IBS schemes [5]. The first approach is a generic transformation from a standard signature scheme introduced by [3] and it is often known as the "certification approach". The second approach is to tightly transform a two-level hierarchical IBE (HIBE) into an IBS scheme [6]. However, neither approach yields short signatures, as the resulting signature either contains a certificate or an HIBE user secret key. Moreover, to obtain an EUF-ID-CMA-secure IBS, the underlying HIBE needs to be adaptively secure, according to [6]. However, the lattice-based HIBE construction achieving adaptive security in the standard model is rather inefficient [7,8]. Such HIBEs are only theoretical results and far from being practical.

An alternative line of research focuses on direct constructions of IBS. There are several IBS schemes in the literature based on bilinear pairings [9–11], where the schemes in [9,10] are proven secure in the random oracle model and the scheme in [11] is secure in the standard model.

In the post-quantum setting, Rückert [12] proposed the first lattice-based hierarchical IBS in both the random oracle and standard models; however, the scheme in the standard model only achieves selective security. In 2021, based on the transformation from EUF-naCMA to EUF-CMA digital signatures, Pan et al. [13] proposed a generic transformation to lift non-adaptively secure IBS schemes to adaptively secure ones. However, the security achieved by their method deviates from the standard EUF-ID-CMA definition and we denote this notion as $\overline{\text{EUF-ID-CMA}}$. In the $\overline{\text{EUF-ID-CMA}}$ security model, the challenger does not store generated signing keys, but instead samples a *fresh* user secret signing key for each $\mathcal{O}_{\mathsf{KeyGen}}$ and $\mathcal{O}_{\mathsf{Sign}}$ query. Furthermore, they did not provide a concrete instantiation of a non-adaptive IBS in the standard model. To date, there is no known direct IBS construction based on standard post-quantum assumptions that achieves EUF-ID-CMA security in the standard model. This motivates the following question:

"Q1: Can we construct a post-quantum IBS scheme that achieves EUF-ID-CMA security in the standard model?"

Identity-Based Matchmaking Encryption. An IB-ME system involves three roles: KGC, senders, and receivers. Each sender or receiver is associated with a unique identity. KGC generates a master secret key and master public key, and delegates the encryption keys for senders and decryption keys for receivers. A sender (with identity) σ uses the corresponding encryption key ek_σ to encrypt a message for a target receiver (with identity) ρ'. A receiver ρ uses the associated decryption key dk_ρ to decrypt a ciphertext from a target sender σ'. A match occurs if the target sender's identity specified by the receiver matches the sender's identity, i.e., $\sigma = \sigma'$, and the target receiver's identity specified by the sender matches the receiver's identity, i.e., $\rho = \rho'$. The security requirements of an IB-ME include privacy and authenticity. Informally, privacy guarantees that a receiver decrypts successfully and learns the message if and only if a match happens, and nothing beyond that is leaked. Authenticity guarantees that any malicious sender cannot pretend to be someone else and forge a valid ciphertext without using the corresponding encryption key.

Ateniese et al. [2] first introduced the IB-ME scheme as a special case of matchmaking encryption (ME). It can also be regarded as a natural generalization of identity-based encryption (IBE). According to [14], IB-ME finds applications in settings where IBE with strong anonymity guarantees is required. In [2] (Subsect. 6.2.2), Ateniese et al. provide a concrete application of IB-ME and show how to implement an anonymous bulletin board hidden service from an IB-ME.

The first construction of IB-ME is due to Ateniese et al. [2] and it is proved to be secure under the bilinear Diffie-Hellman assumption in the random oracle model. Francati et al. [14] subsequently strengthened the privacy notion (and somewhat the authenticity) and gave a generic construction of IB-ME. Although their construction is presented in the plain model, it relies on a non-standard q-augmented bilinear Diffie–Hellman exponent (q-ABDHE) assumption and a non-interactive zero-knowledge (NIZK) proof system. To avoid relying on non-standard (q-type) assumptions, Chen et al. [15] then proposed a more efficient IB-ME scheme from the symmetric external Diffie-Hellman (SXDH) assumption in the standard model. All these IB-ME schemes [2,14,15] were based on bilinear groups and are therefore not post-quantum secure.

In the post-quantum setting, there are several works [16–18] on IB-ME. In [16], Wang et al. proposed a generic construction of IB-ME from HIBE and IBS and additionally provided a lattice-based instantiation in the random oracle model. Notably, Wang et al. [16] introduced the notion of *stronger authenticity* for IB-ME. In this security definition, the adversary is allowed to query an encryption oracle, enabling them to obtain ciphertexts with known plaintexts from chosen sources. Subsequent works [17,18] also proposed IB-ME schemes achieving stronger authenticity via generic frameworks.

However, the security model for stronger authenticity in [16–18] is subtly impractical. Specifically, in their stronger authenticity experiment, the encryption oracle samples a *fresh* encryption key to generate a ciphertext for each query. The issue is that, in practice, a sender typically stores the encryption key and reuses it to generate multiple ciphertexts. This leads to a gap between reality and the security model. To address this gap, we introduce a refined notion of *new stronger authenticity* that captures the reality for IB-ME by allowing the encryption oracle to use the same encryption key for the corresponding identity to generate all ciphertexts. This leads to the following question:

"Q2: Can we construct a post-quantum IB-ME scheme that achieves new stronger authenticity in the standard model?"

1.1. Contribution

In this work, we have made the following contributions.

Construction of IBS from lattices with EUF-ID-CMA security in the standard model. Our IBS instantiation from lattices satisfies EUF-ID-CMA security in the standard model. We show that the security of our IBS scheme reduces to the SIS assumption and the

pseudo-randomness of pseudo-random functions. By employing the LWE-based PRFs in [19], we obtain the first direct lattice-based IBS construction satisfying EUF-ID-CMA security in the standard model, and thus answer question $Q1$.

New stronger authenticity for IB-ME. We introduce the new stronger authenticity notion for IB-ME, which is more practical than the stronger authenticity in [16]. We analyze the limitation of the stronger authenticity in [16] and show that it does not imply our new stronger authenticity by providing a separation.

Generic IB-ME construction with ciphertext pseudo-randomness and new stronger authenticity. We present a generic framework to build IB-ME from an anonymous IBE and an IBS. We prove that the framework's ciphertext pseudo-randomness reduces tightly to the ciphertext pseudo-randomness of the underlying IBE; the framework's new stronger authenticity reduces tightly to the EUF-ID-CMA security of the underlying IBS. By instantiating with our IBS and IBE in [20], we obtain an IB-ME scheme from lattices satisfying new stronger authenticity in the standard model, and thus answer question $Q2$.

1.2. Technique overview

In this subsection, we provide a high-level overview of our IBS and IB-ME constructions.

Instantiation of IBS. Our instantiation of IBS employs PRFs and homomorphic evaluation algorithms $(\mathsf{Eval}_{pub}, \mathsf{Eval}_{prv})$ that support the NC^1 circuit class, following a sequence of works [21–23]. We first introduce some notation. Let $\mathsf{PRF} : \{0,1\}^t \times \{0,1\}^x \to \{0,1\}^y$ be a pseudo-random function with key space $\{0,1\}^t$, input space $\{0,1\}^x$ and output space $\{0,1\}^y$. For vectors $\mathbf{r} \in \{0,1\}^y$ and $\mathbf{x} \in \{0,1\}^x$, define the function $f^{\mathsf{PRF}}_{\mathbf{r},\mathbf{x}} : \{0,1\}^t \to \{0,1\}$ w.r.t. PRF as follows:

$$f^{\mathsf{PRF}}_{\mathbf{r},\mathbf{x}}(\mathbf{k}) = \begin{cases} 1 & \text{if } \mathsf{PRF}(\mathbf{k},\mathbf{x}) = \mathbf{r} \\ 0 & \text{otherwise} \end{cases}$$

According to [21–23], we have two efficient deterministic algorithms Eval_{pub} and Eval_{prv} as follows:

- Given $f^{\mathsf{PRF}}_{\mathbf{r},\mathbf{x}}$ and matrices $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$, $\{\mathbf{A}_i \in \mathbb{Z}_q^{n\times w}\}_{i\in[t]}$, one can homomorphically evaluate a matrix $\mathbf{C}_{f_\mathbf{r}} \in \mathbb{Z}_q^{n\times w}$ by invoking $\mathsf{Eval}_{pub}(f^{\mathsf{PRF}}_{\mathbf{r},\mathbf{x}}(\cdot), \mathbf{A}, \{\mathbf{A}_i\}_{i\in[t]})$.
- Assume that $\mathbf{A}_i = \mathbf{A}\mathbf{R}_i + k_i\mathbf{G}$, where $\mathbf{R}_i \in \mathbb{Z}^{m\times w}$ with small norm, $\mathbf{G}$ is the gadget matrix introduced in [24] and $\mathbf{k} = (k_1, \ldots, k_t)$ is the PRF's key. If additionally $\mathbf{k}$ and $\{\mathbf{R}_i\}_{i\in[t]}$ are given, then one can evaluate a secret matrix $\mathbf{R}_{f_\mathbf{r}} \in \mathbb{Z}^{m\times w}$ by invoking $\mathsf{Eval}_{prv}(f^{\mathsf{PRF}}_{\mathbf{r},\mathbf{x}}(\cdot), \mathbf{A}, \mathbf{k}, \{\mathbf{R}_i\}_{i\in[t]})$. It holds that

$$\mathbf{C}_{f_\mathbf{r}} = \mathbf{A}\mathbf{R}_{f_\mathbf{r}} + f^{\mathsf{PRF}}_{\mathbf{r},\mathbf{x}}(\mathbf{k}) \cdot \mathbf{G} \in \mathbb{Z}_q^{n\times w}.$$

The high-level description of our IBS construction is as follows. The master verification key consists of a matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$, two PRFs, $\widehat{\mathsf{PRF}}$ and $\widetilde{\mathsf{PRF}}$, and matrices $\{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]}$ and $\{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]}$, where $\hat{\mathbf{A}}_i \xleftarrow{\$} \mathbb{Z}_q^{n\times w}$ for $i \in [\hat{t}]$ and $\tilde{\mathbf{A}}_i \xleftarrow{\$} \mathbb{Z}_q^{n\times w}$ for $i \in [\tilde{t}]$. The PRFs $\widehat{\mathsf{PRF}}$ and $\widetilde{\mathsf{PRF}}$ have key lengths $\hat{t}$ and $\tilde{t}$, input lengths $\hat{x}$ and $\tilde{x}$, and output lengths $\hat{y}$ and $\tilde{y}$, respectively. Suppose the message space is $\{0,1\}^\ell$ and the security parameter is κ. We set $\tilde{x} = \hat{x} + \ell + \kappa$. The master secret key consists of a trapdoor $\mathbf{T}_\mathbf{A} \in \mathbb{Z}^{m\times m}$ for $\mathbf{A}$, introduced in [24].

- To generate a signing key for user $\sigma \in \{0,1\}^{\hat{x}}$:
  1. Prepare a matrix $\mathbf{F}_\sigma = [\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}] \in \mathbb{Z}_q^{n\times(m+w)}$, where $\mathbf{C}_{f_{\hat{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f^{\widehat{\mathsf{PRF}}}_{\hat{\mathbf{r}},\sigma}(\cdot), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]})$ with $\hat{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\hat{y}}$.
  2. Delegate a trapdoor $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \in \mathbb{Z}^{(m+w)\times(m+w)}$ for $\mathbf{F}_\sigma$ using $\mathbf{T}_\mathbf{A}$ and set the signing key $sk_\sigma := (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}})$.
- To sign a message $\mathbf{m} \in \{0,1\}^\ell$ for user σ using $sk_\sigma = (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}})$:
  1. Sample $\mathbf{z} \xleftarrow{\$} \{0,1\}^\kappa$ and set $\mathbf{m}' := \mathbf{m}\|\mathbf{z}$.
  2. Prepare a matrix $\mathbf{F}_{\sigma,\mathbf{m}} = [\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}] \in \mathbb{Z}_q^{n\times(m+2w)}$, where $\mathbf{C}_{f_{\hat{\mathbf{r}}}}$ is computed in the same way and $\mathbf{C}_{f_{\tilde{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f^{\widetilde{\mathsf{PRF}}}_{\tilde{\mathbf{r}},\sigma\|\mathbf{m}'}(\cdot), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]})$ with $\tilde{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\tilde{y}}$.
  3. Delegate a trapdoor $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}} \in \mathbb{Z}^{(m+2w)\times(m+2w)}$ for $\mathbf{F}_{\sigma,\mathbf{m}}$ using $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}$.
  4. Sample a short preimage $\mathbf{s}$ of $\mathbf{F}_{\sigma,\mathbf{m}}$ such that $\mathbf{F}_{\sigma,\mathbf{m}} \cdot \mathbf{s} = \mathbf{0}^n$ using $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}$ and set $\Sigma := (\mathbf{s}, \hat{\mathbf{r}}, \tilde{\mathbf{r}}, \mathbf{z})$ as the signature.
- To verify a signature-message tuple $(\Sigma = (\mathbf{s}, \hat{\mathbf{r}}, \tilde{\mathbf{r}}, \mathbf{z}), \mathbf{m})$:
  1. Compute $\mathbf{F}_{\sigma,\mathbf{m}}$ publicly from $\mathbf{A}$, $\{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]}$ and $\{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]}$.
  2. If $\mathbf{F}_{\sigma,\mathbf{m}} \cdot \mathbf{s} = \mathbf{0}^n$, return 1; otherwise, return 0.

Security analysis of the IBS. We first show that the EUF-ID-NMA security (i.e., without the signing oracle) of our IBS reduces tightly to the SIS assumption together with the pseudo-randomness of $\widehat{\mathsf{PRF}}$ and $\widetilde{\mathsf{PRF}}$. The core idea is to embed an SIS challenge into the master verification key and then extract a solution from any successful forgery. Consequently, during the simulation in the security proof we should avoid using the trapdoor $\mathbf{T}_\mathbf{A}$ when answering key generation queries, since $\mathbf{A}$ will be replaced by the SIS instance.

To answer key generation queries despite this replacement, we embed the PRF keys $\hat{\mathbf{k}}$ and $\tilde{\mathbf{k}}$ into the public matrices, namely $\hat{\mathbf{A}}_i = \mathbf{A}\hat{\mathbf{R}}_i + \hat{k}_i \cdot \mathbf{G}$ and $\tilde{\mathbf{A}}_i = \mathbf{A}\tilde{\mathbf{R}}_i + \tilde{k}_i \cdot \mathbf{G}$. For a key generation query on identity σ, we set $\hat{\mathbf{r}} = \widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma)$, so that $f^{\widehat{\mathsf{PRF}}}_{\hat{\mathbf{r}},\sigma}(\hat{\mathbf{k}}) = 1$. Then we have

$$\mathbf{F}_\sigma = [\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}] = [\mathbf{A}|\mathbf{A}\mathbf{R}_{f_{\hat{\mathbf{r}}}} + f^{\widehat{\mathsf{PRF}}}_{\hat{\mathbf{r}},\sigma}(\hat{\mathbf{k}}) \cdot \mathbf{G}] = [\mathbf{A}|\mathbf{A}\mathbf{R}_{f_{\hat{\mathbf{r}}}} + \mathbf{G}],$$

by homomorphic evaluation. Hence $\mathbf{R}_{f_{\hat{\mathbf{r}}}}$ is a gadget trapdoor for $\mathbf{F}_\sigma$ [24], from which we can derive the trapdoor $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}$ required to generate the corresponding private key. Finally, we reduce the EUF-ID-NMA security to the hardness of solving the SIS problem. That is, if there exists an adversary that breaks the EUF-ID-NMA of our IBS by forging a signature on $(\sigma^*, \mathbf{m}^*)$, then we can construct an algorithm $\mathcal{B}$ that solves the SIS problem. In more detail, $\mathcal{B}$ sets the matrix $\mathbf{A}$ as the instance of SIS, and then answers the key generation queries as above. On receiving a forgery $(\sigma^*, \mathbf{m}^*, \Sigma^* = (\mathbf{s}^*, \hat{\mathbf{r}}^*, \tilde{\mathbf{r}}^*, \mathbf{z}^*))$ from $\mathcal{A}$ that passes the verification, it must hold that

$$[\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}^*}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^* = \mathbf{0}^n,$$

where $\mathbf{C}_{f_{\hat{\mathbf{r}}^*}} = \mathbf{A}\mathbf{R}_{f_{\hat{\mathbf{r}}^*}} + f^{\widehat{\mathsf{PRF}}}_{\hat{\mathbf{r}}^*,\sigma^*}(\hat{\mathbf{k}}) \cdot \mathbf{G}$ and $\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}} = \mathbf{A}\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}} + f^{\widehat{\mathsf{PRF}}}_{\tilde{\mathbf{r}}^*,\sigma^*||\mathbf{m}^{*\prime}}(\tilde{\mathbf{k}}) \cdot \mathbf{G}$. Due to the pseudo-randomness of $\widehat{\mathsf{PRF}}$, $\mathcal{A}$ has no knowledge of $\widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma^*)$ and with high probability $f^{\widehat{\mathsf{PRF}}}_{\hat{\mathbf{r}}^*,\sigma^*}(\hat{\mathbf{k}}) = 0$. Then we have $\mathbf{C}_{f_{\hat{\mathbf{r}}^*}} = \mathbf{A}\mathbf{R}_{f_{\hat{\mathbf{r}}^*}}$. Similarly, with high probability, it holds that $\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}} = \mathbf{A}\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}}$. In this case, we have

$$[\mathbf{A}|\mathbf{A}\mathbf{R}_{f_{\hat{\mathbf{r}}^*}}|\mathbf{A}\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^* = \mathbf{A}[\mathbf{I}|\mathbf{R}_{f_{\hat{\mathbf{r}}^*}}|\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^* = \mathbf{0}^n,$$

Since $\mathbf{R}_{f_{\hat{\mathbf{r}}^*}}$, $\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}}$ and $\mathbf{s}^*$ are all of small norms, $[\mathbf{I}|\mathbf{R}_{f_{\hat{\mathbf{r}}^*}}|\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^*$ is a solution to the SIS problem.

However, in the EUF-ID-CMA security proof, we cannot use the same method to generate the trapdoor $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}$ for answering signing queries. Since $\hat{\mathbf{r}}$ is contained in the signature, if $\mathcal{A}$ has queried signatures for the identity σ^*, then it will learn $\hat{\mathbf{r}}^* = \widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma^*)$ from the signature. Consequently, we can no longer ensure that $f^{\widehat{\mathsf{PRF}}}_{\hat{\mathbf{r}}^*,\sigma^*}(\hat{\mathbf{k}}) = 0$ in the forgery. Therefore, we cannot guarantee that $\mathbf{C}_{f_{\hat{\mathbf{r}}^*}} = \mathbf{A}\mathbf{R}_{f_{\hat{\mathbf{r}}^*}}$, and hence we would be unable to extract an SIS solution from it.

A seemingly feasible approach is to compute $\tilde{\mathbf{r}} = \widehat{\mathsf{PRF}}(\tilde{\mathbf{k}}, \sigma\|\mathbf{m}')$ and then use the same method to obtain the trapdoor $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}$, so as to delegate the trapdoor $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}$ with a uniformly chosen $\hat{\mathbf{r}}$. However, this approach does not work, since the adversary can first make a signing query and then ask for the key generation of the same identity σ. It will then find that the two values of $\hat{\mathbf{r}}$ returned in these two queries are different: in the signing query simulation, $\hat{\mathbf{r}}$ is sampled uniformly at random, whereas in the key generation simulation, $\hat{\mathbf{r}}$ is set to $\widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma)$ in order to generate the trapdoor $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}$. Therefore, the adversary can distinguish the simulation from the real game.

Guessing the forgery identity. To bridge the gap between the EUF-ID-NMA proof and the EUF-ID-CMA setting, we guess the target identity σ^* that will appear in the forgery. Let Q_s be an upper bound on the number of signing queries. The reduction algorithm $\mathcal{B}$ proceeds as follows.

1. **Guess the target identity.** At the beginning of the game, $\mathcal{B}$ chooses an index $i^* \leftarrow \{1, \ldots, Q_s + 1\}$ uniformly at random.
   - If $i^* \le Q_s$, $\mathcal{B}$ guesses that the forgery will be on the identity queried in the i^*th signing query. We additionally require that this identity has not been queried to either oracle before; otherwise, the simulation will abort.
   - If $i^* = Q_s + 1$: $\mathcal{B}$ guesses that the forgery will be on some identity σ^* that is never queried to the signing oracle.
2. **Simulating signing queries for the guessed identity.** $\mathcal{B}$ samples $\hat{\mathbf{r}}_{i^*}$ uniformly at random and stores it. Whenever $\mathcal{A}$ requests a signature on $(\sigma_{i^*}, \mathbf{m})$ and $i^* \le Q_s$, $\mathcal{B}$ answers as follows:
   - If this is the i^*th signing query and $\mathcal{A}$ has previously requested the signing key or signature for σ_{i^*}, then $\mathcal{B}$ aborts, since it can no longer keep the responses consistent.
   - Otherwise, $\mathcal{B}$ samples a fresh salt $\mathbf{z} \leftarrow \{0,1\}^\kappa$, sets $\mathbf{m}' = \mathbf{m}\|\mathbf{z}$, and computes $\tilde{\mathbf{r}} = \widehat{\mathsf{PRF}}(\tilde{\mathbf{k}}, \sigma_{i^*}\|\mathbf{m}')$. It then uses the same trapdoor delegation method as in the EUF-ID-NMA proof to obtain a trapdoor for $\mathbf{F}_{\sigma_{i^*},\mathbf{m}} = [\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}]$ and uses it to sample a short preimage $\mathbf{s}$ satisfying $\mathbf{F}_{\sigma_{i^*},\mathbf{m}} \cdot \mathbf{s} = \mathbf{0}^n$.

   The resulting signature $\Sigma = (\mathbf{s}, \hat{\mathbf{r}}_{i^*}, \tilde{\mathbf{r}}, \mathbf{z})$ is computationally indistinguishable from a real signature under the pseudorandomness of PRFs.
3. **Simulating other queries.**
   For any identity $\sigma \neq \sigma_{i^*}$, $\mathcal{B}$ answers key generation queries exactly as in the EUF-ID-NMA proof and answers signing queries by first generating the corresponding key and then producing the signature.
4. **Handling key queries for the guessed identity.** If $\mathcal{A}$ ever requests the signing key for σ_{i^*} (when $i^* \le Q_s$), $\mathcal{B}$ immediately **aborts**. This is necessary because $\mathcal{B}$ cannot produce a valid $sk_{\sigma_{i^*}}$ that is consistent with the already-issued signatures without knowing a trapdoor.
5. **Extracting from the forgery.** Eventually, $\mathcal{A}$ outputs a forgery $(\sigma^*, \mathbf{m}^*, \Sigma^* = (\mathbf{s}^*, \hat{\mathbf{r}}^*, \tilde{\mathbf{r}}^*, \mathbf{z}^*))$.
   - If $i^* \le Q_s$ and $\sigma^* \neq \sigma_{i^*}$, $\mathcal{B}$ **aborts**.
   - If $i^* = Q_s + 1$ and σ^* was ever queried to the signing oracle, $\mathcal{B}$ **aborts**.

   In the remaining case, the guess is correct. Moreover, $\hat{\mathbf{r}}^*$ and $\tilde{\mathbf{r}}^*$ are independent of the corresponding PRF outputs except with negligible probability, so with overwhelming probability we have

   $$f^{\widehat{\mathsf{PRF}}}_{\hat{\mathbf{r}}^*,\sigma^*}(\hat{\mathbf{k}}) = 0 \quad \text{and} \quad f^{\widehat{\mathsf{PRF}}}_{\tilde{\mathbf{r}}^*,\sigma^*\|\mathbf{m}^{*\prime}}(\tilde{\mathbf{k}}) = 0.$$

   Therefore, both $\mathbf{C}_{f_{\hat{\mathbf{r}}^*}}$ and $\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}}$ are of the form $\mathbf{AR}$, and the verification equation yields a valid SIS solution.

Success Probability. The index i^* is chosen uniformly from a set of size $Q_s + 1$ and is hidden from $\mathcal{A}$. Moreover, the abort conditions above are triggered only when the guess i^* is inconsistent with $\mathcal{A}$'s behavior. We will show that the probability that $\mathcal{B}$ does not abort due to a mismatch is exactly $1/(Q_s + 1)$. Conditioned on not aborting and $\mathcal{A}$ producing a valid forgery, $\mathcal{B}$ extracts a valid SIS solution in the EUF-ID-CMA security game.

Construction of IB-ME. Our generic IB-ME framework is built from an IBE with ciphertext pseudo-randomness and an IBS with EUF-ID-CMA security. Both building blocks can be instantiated from the SIS assumption or the LWE assumption in the standard model, yielding the lattice-based IB-ME schemes. We refer readers to Section 4.3 for the detailed description of our generic IB-ME, Section 4.4 for the instantiations of IB-ME from SIS and LWE in the standard model.

More formally, an IB-ME scheme consists of five algorithms IB-ME = (Setup, SKeyGen, RKeyGen, Enc, Dec): a trusted setup algorithm Setup generates the master secret key and master public key (msk, mpk); a sender's key generation algorithm SKeyGen takes as input a sender's identity σ and msk, and generates a sender's encryption key ek_σ; a receiver's key generation algorithm RKeyGen takes as input a receiver's identity ρ and msk, and generates a receiver's decryption key dk_ρ; an encryption algorithm takes as input a sender's encryption key ek_σ, a message m to be encrypted and a target receiver's identity ρ', and returns a ciphertext c; a decryption algorithm takes as input a ciphertext c, a receiver's decryption key dk_ρ and a target sender's identity σ', and returns the plaintext if a match happens and $\perp$ otherwise. An IBE scheme consists of four algorithms IBE = (IBE.Setup, IBE.KeyGen, IBE.Enc, IBE.Dec). Similarly, an IBS scheme consists of four algorithms IBS = (IBS.Setup, IBS.KeyGen, IBS.Sign, IBS.Ver).

Our generic IB-ME follows the sign-then-encrypt construction. We set $mpk = (\mathsf{IBE}.mpk, \mathsf{IBS}.mvk)$ and $msk = (\mathsf{IBE}.msk, \mathsf{IBS}.msk)$. The sender's encryption key ek_σ is the signing key obtained by invoking $\mathsf{IBS}.sk_\sigma \leftarrow \mathsf{IBS.KeyGen}(\mathsf{IBS}.msk, \sigma)$. The receiver's decryption key dk_ρ is the decryption key obtained by invoking $\mathsf{IBE}.sk_\rho \leftarrow \mathsf{IBE.KeyGen}(\mathsf{IBE}.msk, \rho)$. To encrypt a message m for target receiver ρ, a sender σ first computes a signature Σ of (σ, ρ, m) by invoking $\Sigma \leftarrow \mathsf{IBS.Sign}(\mathsf{IBS}.sk_\sigma, (\sigma, \rho, m))$ with the corresponding encryption key ek_σ, and then computes the ciphertext c on message (Σ, m) by invoking $c \leftarrow \mathsf{IBE.Enc}(\mathsf{IBE}.mpk, \rho, (\Sigma, m))$. To decrypt a ciphertext c, a receiver ρ first computes $(\Sigma, m) \leftarrow \mathsf{IBE.Dec}(\mathsf{IBE}.mpk, \mathsf{IBE}.sk_\rho, c)$ and returns m if $\mathsf{IBS.Ver}(\mathsf{IBS}.mvk, \sigma, (\sigma, \rho, m), \Sigma) = 1$ for target sender σ.

The ciphertext pseudo-randomness of our IB-ME follows from the ciphertext pseudo-randomness of the underlying IBE. Specifically, if ρ is not the target receiver's identity specified in c, then the ciphertext is pseudo-random and the receiver obtains nothing about the plaintext and the sender's identity. The new stronger authenticity of our IB-ME follows from the EUF-ID-CMA security of the underlying IBS.

2. Preliminaries

2.1. Notations

In this paper, column vectors are denoted by bold lower-case letters like $\mathbf{x}$ and the ith component of $\mathbf{x}$ is denoted by x_i. Matrices are denoted by bold upper-case letters like $\mathbf{A}$ and the ith column of $\mathbf{A}$ is denoted by $\mathbf{a}_i$. The transpose of $\mathbf{A}$ is denoted by $\mathbf{A}^\top$.

For an integer $n \in \mathbb{N}$, let $[n]$ denote the finite set $\{1, 2, \ldots, n\}$. For a list $\mathcal{L}$ with n entries, denote by $\mathcal{L}[i]$ the ith entry of $\mathcal{L}$ for $i \le n$.

Let κ denote the security parameter and $\mathsf{poly}(\kappa)$ denote a polynomial function. Let $\mathsf{negl} : \mathbb{N} \to \mathbb{R}$ denote a negligible function, i.e., for any polynomial $\mathsf{poly}(n)$, there exists an $n' \in \mathbb{N}$ s.t., for all $n > n'$, $\mathsf{negl}(n) < 1/\mathsf{poly}(n)$. For a primitive XX and a security notion YY, we denote by $\mathsf{Exp}^{\mathsf{YY}}_{\mathsf{XX},\mathcal{A}}(\kappa) \Rightarrow b$ a security experiment interacting with an adversary $\mathcal{A}$ and returning a bit b. Furthermore, we denote by $\mathsf{Adv}^{\mathsf{YY}}_{\mathsf{XX},\mathcal{A}}(\kappa)$ the advantage of $\mathcal{A}$ in $\mathsf{Exp}^{\mathsf{YY}}_{\mathsf{XX},\mathcal{A}}(\kappa)$. Here PPT denotes probabilistic polynomial time.

For a distribution (or a random variable) X, let $x \leftarrow X$ denote the process of sampling x according to X. For a finite set $\mathcal{X}$, let $x \xleftarrow{\$} \mathcal{X}$ denote the process of sampling x from the uniform distribution over $\mathcal{X}$. Let X and Y be two distributions over support S, then the statistical distance between X and Y is defined by $\mathsf{SD}(X, Y) = 1/2 \cdot \sum_{s \in S} |\Pr[X = s] - \Pr[Y = s]|$. We use $\$$ to denote a uniform distribution over some proper set defined in the context.

For a randomized algorithm Alg, we denote by $\mathsf{Alg}(\mathsf{input}; \mathsf{random})$ the process of invoking Alg on the input input using the randomness random and denote by $\mathcal{T}(\mathsf{Alg})$ the running time of Alg.

2.2. Lattice background

Let k, n, m, q be positive integers. Given k linearly independent basis vectors $\mathbf{a}_1, \ldots, \mathbf{a}_k \in \mathbb{R}^n$, let $\mathbf{A} := (\mathbf{a}_1, \ldots, \mathbf{a}_k)$. Then the k-dimensional lattice $\Lambda(\mathbf{A})$ generated by $\mathbf{A}$ is $\Lambda(\mathbf{A}) = \{\mathbf{y} \in \mathbb{R}^n | \mathbf{y} = \mathbf{A}\mathbf{x}, \mathbf{x} \in \mathbb{Z}^k\}$. For an integer q, matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and $\mathbf{u} \in \mathbb{Z}_q^n$, define the q-ary lattice $\Lambda_q(\mathbf{A}) := \{\mathbf{A}^\top \mathbf{x} | \mathbf{x} \in \mathbb{Z}_q^n\} + q\mathbb{Z}^m$, and its "orthogonal" lattice $\Lambda_q^\perp(\mathbf{A}) := \{\mathbf{x} \in \mathbb{Z}^m | \mathbf{A}\mathbf{x} = \mathbf{0}^n \mod q\}$ and "shifted" lattice $\Lambda_q^{\mathbf{u}}(\mathbf{A}) := \{\mathbf{x} \in \mathbb{Z}^m | \mathbf{A}\mathbf{x} = \mathbf{u} \mod q\}$.

The Euclidean norm of a vector $\mathbf{v}$ (resp. a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$) is denoted by $\|\mathbf{v}\|$ (resp. $\|\mathbf{A}\| = \max_{i \in [m]} \|\mathbf{a}_i\|$), the maximum singular value of a matrix $\mathbf{A}$ is defined by $s_1(\mathbf{A}) = \max_{\|\mathbf{x}\|=1} \|\mathbf{A}\mathbf{x}\|$, the Gram-Schmidt orthogonalization of a matrix $\mathbf{A}$ is denoted by $\tilde{\mathbf{A}}$.

Definition 1 (Discrete Gaussian distribution)**.** The Gaussian function with parameter s and center $\mathbf{c} \in \mathbb{R}^n$ is defined as $\rho_{s,\mathbf{c}} : \mathbb{R}^n \to \mathbb{R}$, $\rho_{s,\mathbf{c}}(\mathbf{x}) := \exp(-\pi \|\mathbf{x} - \mathbf{c}\|^2 / s^2)$. For a countable set $S \subset \mathbb{R}^n$, the discrete Gaussian distribution $D_{S,s,\mathbf{c}}$ with parameter s and center $\mathbf{c}$ is defined as $D_{S,s,\mathbf{c}}(\mathbf{x}) := \rho_{s,\mathbf{c}}(\mathbf{x}) / \sum_{\mathbf{x} \in S} \rho_{s,\mathbf{c}}(\mathbf{x})$ for $\mathbf{x} \in S$ and $D_{S,s,\mathbf{c}}(\mathbf{x}) := 0$ for $\mathbf{x} \notin S$. Usually, s is omitted when $s = 1$ and $\mathbf{c}$ is omitted if $\mathbf{c} = \mathbf{0}$.

Lemma 1 (Trapdoor generation [25])**.** *Let q, n, m be positive parameters s.t., q is odd, $q \ge 3$ and $m = O(n \log q)$. There exists a PPT algorithm* $\mathsf{TrapGen}(1^n, 1^m, q)$ *that outputs matrices $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and $\mathbf{T}_\mathbf{A} \in \mathbb{Z}^{m \times m}$ s.t., the distribution of $\mathbf{A}$ is statistically close to a uniform rank n matrix*

in $\mathbb{Z}_q^{n\times m}$ and matrix $\mathbf{T_A}$ is a trapdoor for $\mathbf{A}$ (i.e., a basis for $\Lambda^{\perp}(\mathbf{A})$) satisfying $\|\tilde{\mathbf{T}}_{\mathbf{A}}\| \le O(\sqrt{n\log q})$ and $\|\mathbf{T_A}\| \le O(n\log q)$ with all but 2^{-n} probability.

Lemma 2 (Preimage sampling [26]). *Let q, n, m, δ be positive parameters s.t., $q \ge 2$. Let $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ be a matrix with a trapdoor $\mathbf{T_A} \in \mathbb{Z}^{m\times m}$. Let $\delta \ge \|\tilde{\mathbf{T}}_{\mathbf{A}}\| \cdot \omega(\sqrt{\log m})$. Let $\mathbf{u} \in \mathbb{Z}_q^n$ be some vector. There exists a PPT algorithm $\mathsf{SamplePre}(\mathbf{A}, \mathbf{T_A}, \mathbf{u}, \delta)$ that outputs a vector $\mathbf{s} \in \mathbb{Z}^m$ sampled from a distribution statistically close to $D_{\Lambda_q^{\mathbf{u}}(\mathbf{A}),\delta}$.*

Lemma 3 (Trapdoor delegation [27]). *Let $q, n, m, \bar{m}, m'$ be positive parameters and $m' = m + \bar{m}$. Let $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ and $\bar{\mathbf{A}} \in \mathbb{Z}_q^{n\times \bar{m}}$ be matrices and $\mathbf{T_A} \in \mathbb{Z}^{m\times m}$ be a trapdoor for $\mathbf{A}$. There exists a deterministic polynomial-time algorithm $\mathsf{DelTrap}(\mathbf{A}' = [\mathbf{A}|\bar{\mathbf{A}}], \mathbf{T_A})$ that outputs a trapdoor $\mathbf{T}_{\mathbf{A}|\bar{\mathbf{A}}} \in \mathbb{Z}^{m'\times m'}$ for the matrix $\mathbf{A}'$. Besides, it holds that $\|\tilde{\mathbf{T}}_{\mathbf{A}'}\| = \|\tilde{\mathbf{T}}_{\mathbf{A}}\|$.*

According to [27], the columns of $\mathbf{A}'$ can be permuted in any order; this only permutes the coordinates of vectors in $\Lambda^{\perp}(\mathbf{A}')$ accordingly. Formally, for any permutation matrix $\mathbf{P} \in \{0,1\}^{m'\times m'}$, we have $\Lambda_q^{\perp}(\mathbf{A}'\mathbf{P}) = \mathbf{P}^{\top} \cdot \Lambda_q^{\perp}(\mathbf{A}')$. Specifically, let $\mathbf{A}' = [\mathbf{A}_1|\mathbf{A}_2|\mathbf{A}_3]$. If we already have a trapdoor $\mathbf{T}_{[\mathbf{A}_1|\mathbf{A}_3]}$ for $[\mathbf{A}_1|\mathbf{A}_3]$, then by computing $\mathbf{T}_{[\mathbf{A}_1|\mathbf{A}_3|\mathbf{A}_2]} \leftarrow \mathsf{DelTrap}([\mathbf{A}_1|\mathbf{A}_3|\mathbf{A}_2], \mathbf{T}_{[\mathbf{A}_1|\mathbf{A}_3]})$ and applying the corresponding permutation, one obtains a trapdoor for $\mathbf{A}'$.

Lemma 4 (Basis randomization [27]). *Let q, n, m, γ be positive parameters. Let $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ be a matrix with a trapdoor $\mathbf{T_A} \in \mathbb{Z}^{m\times m}$. Let $\gamma \ge \|\tilde{\mathbf{T}}_{\mathbf{A}}\| \cdot \omega(\sqrt{\log n})$. There exists a PPT algorithm $\mathsf{RandBasis}(\mathbf{T_A}, \gamma)$ that outputs a new basis $\mathbf{T}'_{\mathbf{A}}$ for $\mathbf{A}$ s.t., $\|\tilde{\mathbf{T}}'_{\mathbf{A}}\| \le \gamma \cdot \sqrt{m}$ with overwhelming probability. Besides, the distribution of $\mathbf{T}'_{\mathbf{A}}$ is statistically close to $\mathsf{RandBasis}(\mathbf{T}^*_{\mathbf{A}}, \gamma)$ for any other trapdoor $\mathbf{T}^*_{\mathbf{A}}$ of $\mathbf{A}$ satisfying $\gamma \ge \|\tilde{\mathbf{T}}^*_{\mathbf{A}}\| \cdot \omega(\sqrt{\log n})$.*

In this paper, we also use the G-trapdoor defined by Micciancio and Peikert in [24]. Let q, n, m be integers and define $w = n\lceil \log q \rceil$. Then a G-trapdoor of a matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ is a matrix $\mathbf{R} \in \mathbb{Z}^{(m-w)\times w}$ s.t., $\mathbf{A} \cdot [-\mathbf{R}^{\top}|\mathbf{I}_w^{\top}]^{\top} = \mathbf{G}$, where $\mathbf{G}$ is the gadget matrix. Given matrices $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ and $\mathbf{R} \in \mathbb{Z}^{m\times w}$, it is easy to see that $\mathbf{R}$ is a G-trapdoor of matrix $[\mathbf{A}|\mathbf{AR} + \mathbf{G}]$. We recall the following Definition 2 and Lemma 5 from [24].

Definition 2 (Gadget matrix [24]). For any integer modulus q, the gadget vector over $\mathbb{Z}_q$ is defined as $\mathbf{g}^{\top} := (1, 2, 4, \ldots, 2^{\lceil \log q \rceil - 1}) \in \mathbb{Z}_q^{1\times \lceil \log q \rceil}$. Let $w = n\lceil \log q \rceil$, the gadget matrix $\mathbf{G}$ with full row rank is defined as:

$$\mathbf{G} = \mathbf{I}_n \otimes \mathbf{g}^{\top} = \begin{pmatrix} \mathbf{g}^{\top} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{g}^{\top} & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{g}^{\top} \end{pmatrix} \in \mathbb{Z}_q^{n\times w}.$$

Lemma 5 (G-to-Basis [24]). *Let n, m, q be positive integers and define $w = n\lceil \log q \rceil$. Let $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ be a matrix with a G-trapdoor $\mathbf{R} \in \mathbb{Z}^{(m-w)\times w}$. There exists a PPT algorithm $\mathsf{GtoBasis}(\mathbf{R})$ that returns a trapdoor $\mathbf{T_A}$ of $\mathbf{A}$ (i.e., a basis for $\Lambda^{\perp}(\mathbf{A})$). Moreover, the trapdoor $\mathbf{T_A}$ satisfies that $\|\tilde{\mathbf{T}}_{\mathbf{A}}\| \le \sqrt{5}(s_1(\mathbf{R}) + 1)$.*

Lemma 6 (Randomness extraction [28]). *Let q, n, m be positive integers s.t., $q > 2$ is a prime and $m \ge (n+1)\log q + \omega(\log n)$. If $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$, $\mathbf{R} \xleftarrow{\$} \{1, -1\}^{m\times w}$ and $\mathbf{U} \xleftarrow{\$} \mathbb{Z}_q^{n\times w}$, then $\mathsf{SD}((\mathbf{A}, \mathbf{AR}), (\mathbf{A}, \mathbf{U})) \le \mathsf{negl}(n)$.*

Lemma 7 (Homomorphic evaluation [20–23,29–31]). *Let q, n, m, ℓ, k be positive integers and define $w = n\lceil \log q \rceil$. Let $\mathbf{G} \in \mathbb{Z}_q^{n\times w}$ be the gadget matrix. Given a NAND boolean circuit $C : \{0,1\}^{\ell} \to \{0,1\}^k$ with circuit depth d, vector $\mathbf{x} = (x_1, \ldots, x_{\ell})^{\top} \in \{0,1\}^{\ell}$, and matrices $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$, $(\mathbf{A}_i \in \mathbb{Z}_q^{n\times w})_{i\in[\ell]}$ and $(\mathbf{R}_i \in \{\pm 1\}^{m\times w})_{i\in[\ell]}$, the following holds:*

- *There is an efficient deterministic algorithm $\mathsf{Eval}_{pub}(C, \mathbf{A}, (\mathbf{A}_i)_{i\in[\ell]})$ that takes as input the circuit C and matrices $\mathbf{A}$, $(\mathbf{A}_i)_{i\in[\ell]}$, and outputs a matrix $\mathbf{A}_C \in \mathbb{Z}_q^{n\times kw}$.*
- *There is an efficient deterministic algorithm $\mathsf{Eval}_{prv}(C, \mathbf{A}, \mathbf{x}, (\mathbf{R}_i)_{i\in[\ell]})$ that takes as input the circuit C, matrix $\mathbf{A}$, vector $\mathbf{x}$ and matrices $(\mathbf{R}_i)_{i\in[\ell]}$, and outputs a matrix $\mathbf{R}_C \in \mathbb{Z}^{m\times kw}$.*

Homomorphism. *If $\mathbf{A}_i = \mathbf{AR}_i + x_i \cdot \mathbf{G} \in \mathbb{Z}_q^{n\times w}$ for all $i \in [\ell]$, $\mathbf{A}_C \leftarrow \mathsf{Eval}_{pub}(C, \mathbf{A}, (\mathbf{A}_i)_{i\in[\ell]})$ and $\mathbf{R}_C \leftarrow \mathsf{Eval}_{prv}(C, \mathbf{A}, \mathbf{x}, (\mathbf{R}_i)_{i\in[\ell]})$, then we have*

$$\mathbf{A}_C = \mathbf{AR}_C + C(\mathbf{x}) \otimes \mathbf{G},$$

where $s_1(\mathbf{R}_C) \le O(4^d \cdot m^{\frac{3}{2}})$. Particularly, when C is in the circuit class NC^1, i.e., C has depth $d = c\log \ell$ for some constant c, we have $s_1(\mathbf{R}_C) \le O(\ell^{2c} \cdot m^{\frac{3}{2}})$.

Definition 3 (The SIS assumption). Let q, n, m be integers and β be a positive real. The (homogeneous) small integer solution (SIS) assumption $\mathsf{SIS}_{n,q,\beta,m}$ states that for any PPT adversary $\mathcal{A}$, its advantage satisfies:

$$\mathsf{Adv}^{\mathsf{SIS}}_{[n,q,\beta,m],\mathcal{A}}(\kappa) := \Pr\left[\mathcal{A}(\mathbf{A}) \to \mathbf{e} : \mathbf{Ae} = \mathbf{0}^n \ \wedge \mathbf{e} \ne \mathbf{0}^m \wedge \ \|\mathbf{e}\| \le \beta\right] \le \mathsf{negl}(\kappa),$$

where $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$ and $\mathbf{0}^n = (0, \ldots, 0)^{\top} \in \mathbb{Z}_q^n$.

Lemma 8 (The hardness of SIS [26]). *For any $m, \beta = \mathsf{poly}(n)$ and for any prime $q \ge \beta \cdot \omega(\sqrt{n\log n})$, solving the average-case problem $\mathsf{SIS}_{n,q,\beta,m}$ with non-negligible probability is as hard as approximating the shortest independent vectors problem SIVP_{γ} in the worst case, for certain $\gamma = \beta \cdot \widetilde{O}(\sqrt{n})$ factors.*

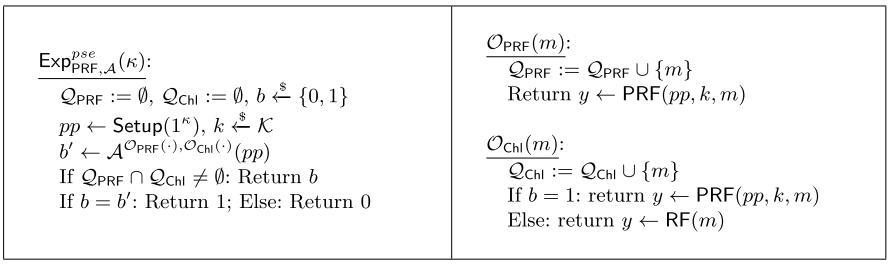


Fig. 1. Experiment $\mathsf{Exp}^{pse}_{\mathsf{PRF},\mathcal{A}}$ for defining the pseudo-randomness of PRF.

2.3. Pseudo-random function

Definition 4 (Pseudo-random function family [32])**.** A pseudo-random function family $\mathsf{PRF} := \{F : \mathcal{K} \times \mathcal{X} \to \mathcal{Y}\}$ is equipped with two polynomial time algorithms $(\mathsf{Setup}, \mathsf{PRF})$ defined below.

- $\mathsf{Setup}(1^\kappa)$ takes as input the security parameter $\kappa \in \mathbb{N}$ and outputs a public parameter pp.
- $\mathsf{PRF}(pp, k, x)$ takes as input the public parameter pp, key $k \in \mathcal{K}$ and message $x \in \mathcal{X}$, and outputs $y \in \mathcal{Y}$. For simplicity, we will omit pp and just write it as $\mathsf{PRF}(k, x)$ when the context is clear.

Pseudo-randomness. Let $\mathsf{RF} : \mathcal{X} \to \mathcal{Y}$ be a truly random function. For any PPT adversary $\mathcal{A}$, its advantage satisfies

$$\mathsf{Adv}^{pse}_{\mathsf{PRF},\mathcal{A}}(\kappa) := \left| \Pr[\mathsf{Exp}^{pse}_{\mathsf{PRF},\mathcal{A}}(\kappa) \Rightarrow 1] - 1/2 \right| \leq \mathsf{negl}(\kappa),$$

where the experiment $\mathsf{Exp}^{pse}_{\mathsf{PRF},\mathcal{A}}$ is defined in Fig. 1.

Note that the pseudo-randomness notion defined in Fig. 1 is different from the standard one considered for a PRF. For the standard pseudo-randomness, the adversary is only given the challenge oracle $\mathcal{O}_{\mathsf{Chl}}$. As for our definition, the adversary is allowed to have additional access to a real PRF oracle $\mathcal{O}_{\mathsf{PRF}}$, which always returns real PRF values. Below we show that the two definitions tightly imply each other and present the following Theorem 1. The proof of Theorem 1 is given in Appendix A.

Theorem 1. *The standard pseudo-randomness and the pseudo-randomness defined in Fig. 1 tightly imply each other. More specifically, for any PPT adversary $\mathcal{A}_1$ against the pseudo-randomness and adversary $\mathcal{A}_2$ against the standard pseudo-randomness (abbr. std-pse), there exist PPT adversaries $\mathcal{B}_1$ and $\mathcal{B}_2$ s.t.*

$$\mathsf{Adv}^{pse}_{\mathsf{PRF},\mathcal{A}_1}(\kappa) \leq 2 \cdot \mathsf{Adv}^{std\text{-}pse}_{\mathsf{PRF},\mathcal{B}_1}(\kappa),\ \mathsf{Adv}^{std\text{-}pse}_{\mathsf{PRF},\mathcal{A}_2}(\kappa) \leq \mathsf{Adv}^{pse}_{\mathsf{PRF},\mathcal{B}_2}(\kappa),$$

where $\mathcal{T}(\mathcal{B}_1) \approx \mathcal{T}(\mathcal{A}_1)$ and $\mathcal{T}(\mathcal{B}_2) \approx \mathcal{T}(\mathcal{A}_2)$.

An almost tight PRF with standard pseudo-randomness from the LWE assumption can be found in [19], and we denote it by $\mathsf{PRF}_{\mathsf{BPR}}$. According to [19], $\mathsf{PRF}_{\mathsf{BPR}}$ can be implemented in the NC^1 circuit class via preprocessing. Together with our Theorem 1, the following Lemma 9 holds.

Lemma 9. *Let n, m, q, k be positive parameters and θ be the parameter for a Gaussian distribution, and $k(\theta\sqrt{n})^k \cdot n^{\omega(1)} \leq q \leq 2^{\mathsf{poly}(n)}$. There exists a PRF $\mathsf{PRF}_{\mathsf{BPR}}$ with input space $\{0,1\}^k$ that satisfies pseudo-randomness. More precisely, for any PPT adversary $\mathcal{A}$ against the pseudo-randomness of $\mathsf{PRF}_{\mathsf{BPR}}$, there exists PPT algorithm $\mathcal{B}$ s.t.,*

$$\mathsf{Adv}^{pse}_{\mathsf{PRF}_{\mathsf{BPR}},\mathcal{A}}(\kappa) \leq 2 \cdot \mathsf{Adv}^{std\text{-}pse}_{\mathsf{PRF}_{\mathsf{BPR}},\mathcal{B}}(\kappa),$$

where $\mathcal{T}(\mathcal{B}) \approx \mathcal{T}(\mathcal{A})$.

2.4. Lamport one-time signature

We recall the Lamport one-time signature scheme from [33]. Let f be a one-way function with input length κ and let ℓ denote the message length. The Lamport one-time signature scheme $\mathsf{Sig}_{\mathsf{lam}} = (\mathsf{KeyGen}, \mathsf{Sign}, \mathsf{Ver})$ is shown below.

- $\mathsf{KeyGen}(1^\kappa)$ takes as input the security parameter κ. For each $i \in [\ell]$ and $b \in \{0,1\}$, sample $x_{i,b} \xleftarrow{\$} \{0,1\}^\kappa$ and compute $y_{i,b} = f(x_{i,b})$. Finally, it returns the secret signing key $sk := \begin{pmatrix} x_{1,0} & x_{2,0} & \cdots & x_{\ell,0} \\ x_{1,1} & x_{2,1} & \cdots & x_{\ell,1} \end{pmatrix}$ and public verification key $vk := \begin{pmatrix} y_{1,0} & y_{2,0} & \cdots & y_{\ell,0} \\ y_{1,1} & y_{2,1} & \cdots & y_{\ell,1} \end{pmatrix}$.
- $\mathsf{Sign}(sk, m)$ takes as input a secret signing key $sk = \begin{pmatrix} x_{1,0} & x_{2,0} & \cdots & x_{\ell,0} \\ x_{1,1} & x_{2,1} & \cdots & x_{\ell,1} \end{pmatrix}$ and a message $m = m_1 \cdots m_\ell \in \{0,1\}^\ell$, returns the signature $\sigma = (x_{1,m_1}, x_{2,m_2}, \cdots, x_{\ell,m_\ell})$.

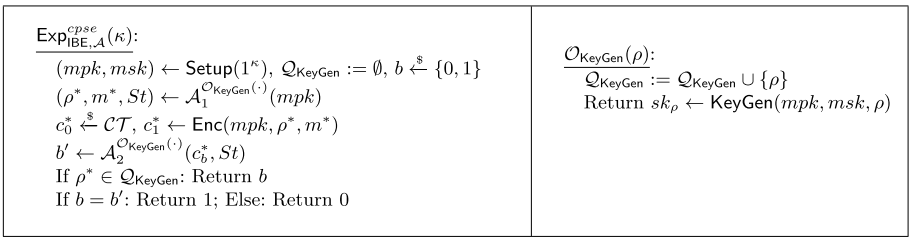


Fig. 2. Experiment $\mathsf{Exp}^{cpse}_{\mathsf{IBE},\mathcal{A}}$ for defining the ciphertext pseudo-randomness of IBE.

- $\mathsf{Ver}(vk, m, \sigma)$ takes as input a public verification key $vk = \begin{pmatrix} y_{1,0} & y_{2,0} & \cdots & y_{\ell,0} \\ y_{1,1} & y_{2,1} & \cdots & y_{\ell,1} \end{pmatrix}$, a message $m = m_1 \cdots m_\ell$ and a signature $\sigma = (x_1, x_2, \ldots, x_\ell)$. It returns 1 if $f(x_i) = y_{i,m_i}$ for all $i \in [\ell]$ and 0 otherwise.

Lemma 10. *Let* $\mathsf{Sig}_{\mathsf{lam}}$ *be the Lamport one-time signature scheme instantiated with a one-way function* f *and message length* ℓ*. Then* $\mathsf{Sig}_{\mathsf{lam}}$ *is existentially unforgeable under one-time chosen-message attack (EUF-OT-CMA).*

Fact 1. *The Lamport one-time signature scheme* $\mathsf{Sig}_{\mathsf{lam}}$ *is NOT EUF-CMA secure. In fact, we can easily present an EUF-CMA attack: Let* $m = M_1 \| M_2$ *and* $m' = M_1 \| \overline{M}_2$*, where* $M_1 \in \{0,1\}^{\ell_1}$*,* $M_2 \in \{0,1\}^{\ell-\ell_1}$*, and* $\overline{M}_2 \in \{0,1\}^{\ell-\ell_1}$ *denotes the bit-wise complement of* M_2*. After issuing two queries* m *and* m' *to the signing oracle, the adversary is able to obtain two signatures* $\sigma = (x_{1,m_1}, \ldots, x_{\ell_1,m_{\ell_1}}, x_{\ell_1+1,m_{\ell_1+1}}, \ldots, x_{\ell,m_\ell})$ *and* $\sigma' = (x_{1,m'_1}, \ldots, x_{\ell_1,m'_{\ell_1}}, x_{\ell_1+1,m'_{\ell_1+1}}, \ldots, x_{\ell,m'_\ell})$*. Consequently, it recovers* $x_{1,m_1}, \ldots, x_{\ell_1,m_{\ell_1}}$ *and both* $x_{i,0}$ *and* $x_{i,1}$ *for each* $i \in [\ell_1 + 1, \ell]$*. Therefore it can forge a valid signature* $\sigma^* = (x_{1,m_1^*}, \ldots, x_{\ell_1,m^*_{\ell_1}}, x_{\ell_1+1,m^*_{\ell_1+1}}, \ldots, x_{\ell,m^*_\ell})$ *for any message* $M^* = M_1 \| M_2^*$ *for* $M_2^* \in \{0,1\}^{\ell-\ell_1}$*.*

2.5. Identity-based encryption

Definition 5 (Identity-based encryption)**.** Let $\mathcal{ID}$ be the identity space, $\mathcal{M}$ be the message space and $\mathcal{CT}$ be the ciphertext space. An identity-based encryption (IBE) scheme IBE consists of four polynomial-time algorithms (Setup, KeyGen, Enc, Dec) defined below.

- $\mathsf{Setup}(1^\kappa)$ takes as input the security parameter κ, and returns the master public key mpk and master secret key msk.
- $\mathsf{KeyGen}(mpk, msk, \rho)$ takes as input the master public key mpk, the master secret key msk and a user's identity $\rho \in \mathcal{ID}$, and returns a private key sk_ρ.
- $\mathsf{Enc}(mpk, \rho, m)$ takes as input the master public key mpk, a target user's identity $\rho \in \mathcal{ID}$ and a message $m \in \mathcal{M}$, and returns a ciphertext c.
- $\mathsf{Dec}(mpk, sk_\rho, c)$ takes as input the master public key mpk, a private key sk_ρ and a ciphertext c, and returns a message m or $\perp$.

Correctness. For all $\kappa \in \mathbb{N}$, $(mpk, msk) \leftarrow \mathsf{Setup}(1^\kappa)$, identity $\rho \in \mathcal{ID}$, $sk_\rho \leftarrow \mathsf{KeyGen}(mpk, msk, \rho)$, $m \in \mathcal{M}$, it holds that:

$$\Pr\left[\mathsf{Dec}(mpk, sk_\rho, \mathsf{Enc}(mpk, \rho, m)) = m\right] \geq 1 - \mathsf{negl}(\kappa).$$

Ciphertext pseudo-randomness [34].[1] An IBE scheme IBE satisfies the ciphertext pseudo-randomness if for any PPT adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, its advantage satisfies:

$$|\Pr[\mathsf{Exp}^{cpse}_{\mathsf{IBE},\mathcal{A}}(\kappa) \Rightarrow 1] - 1/2| \leq \mathsf{negl}(\kappa),$$

where the security experiment $\mathsf{Exp}^{cpse}_{\mathsf{IBE},\mathcal{A}}$ is defined in Fig. 2.

3. Identity-based signature from lattices

3.1. IBS: syntax and EUF-ID-CMA security

Definition 6 (Identity-based signature)**.** Let $\mathcal{ID}$ be an identity space and $\mathcal{M}$ be a message space. An identity-based signature scheme IBS consists of four polynomial-time algorithms (Setup, KeyGen, Sign, Ver) defined below.

- $\mathsf{Setup}(1^\kappa)$ takes as input the security parameter κ, and returns the master verification key mvk and master secret key msk.
- $\mathsf{KeyGen}(msk, \sigma)$ takes as input the master secret key msk and a user's identity $\sigma \in \mathcal{ID}$, and returns a secret signing key sk_σ w.r.t. σ.
- $\mathsf{Sign}(sk_\sigma, m)$ takes as input the secret signing key sk_σ and a message $m \in \mathcal{M}$, and returns a signature Σ.
- $\mathsf{Ver}(mvk, \sigma, m, \Sigma)$ takes as input the master verification key mvk, an identity $\sigma \in \mathcal{ID}$, a message $m \in \mathcal{M}$ and a signature Σ, and returns a decision bit $b \in \{0,1\}$.

[1] This notion is termed adaptive anonymity in [34].

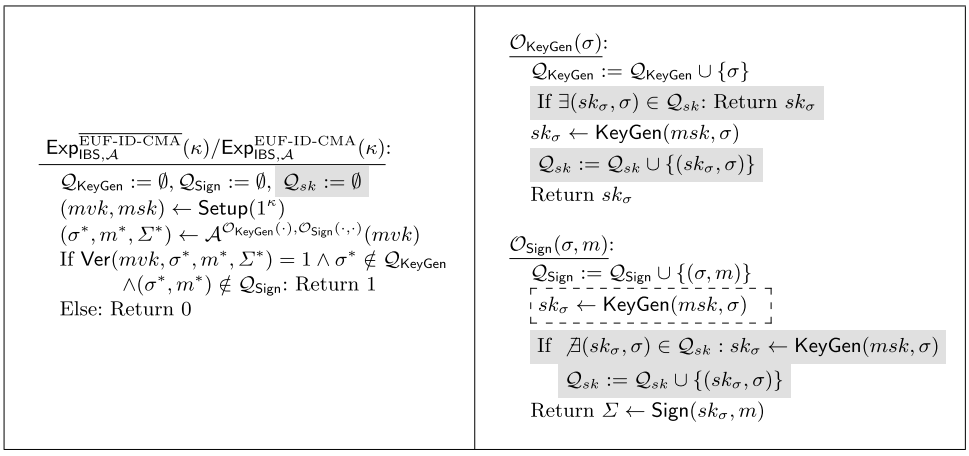


Fig. 3. Experiment $\mathsf{Exp}^{\overline{\text{EUF-ID-CMA}}}_{\mathsf{IBS},\mathcal{A}}$ (with the dashed parts) adopted from [13,16] and Experiment $\mathsf{Exp}^{\text{EUF-ID-CMA}}_{\mathsf{IBS},\mathcal{A}}$ (with the gray parts) for defining unforgeability for IBS refined from [3,4].

Correctness. For all $\kappa \in \mathbb{N}$ and all $m \in \mathcal{M}$, if $(mvk, msk) \leftarrow \mathsf{Setup}(1^\kappa)$, $\sigma \in \mathcal{ID}$, $sk_\sigma \leftarrow \mathsf{KeyGen}(msk, \sigma)$ and $\Sigma \leftarrow \mathsf{Sign}(sk_\sigma, m)$, then

$$\Pr[\mathsf{Ver}(mvk, \sigma, m, \Sigma) = 1] \geq 1 - \mathsf{negl}(\kappa).$$

The standard security notion for identity-based signatures (IBS) is existential unforgeability under chosen-identity and chosen-message attacks (EUF-ID-CMA). However, there are two slightly different definitions of EUF-ID-CMA security for IBS in the literature. The version in [3,4] is the most widely adopted, while an alternative definition, denoted here by $\overline{\text{EUF-ID-CMA}}$, is given in [13,16].

We first recall both definitions and then highlight the difference between them.

EUF-ID-CMA [3,4]. An IBS scheme IBS is EUF-ID-CMA secure if for any PPT adversary $\mathcal{A}$, its advantage satisfies

$$\Pr\left[\mathsf{Exp}^{\text{EUF-ID-CMA}}_{\mathsf{IBS},\mathcal{A}}(\kappa) \Rightarrow 1\right] \leq \mathsf{negl}(\kappa),$$

where the security experiment $\mathsf{Exp}^{\text{EUF-ID-CMA}}_{\mathsf{IBS},\mathcal{A}}$ is defined in Fig. 3.

$\overline{\text{EUF-ID-CMA}}$ [13,16]. An IBS scheme IBS is $\overline{\text{EUF-ID-CMA}}$ secure if for any PPT adversary $\mathcal{A}$, its advantage satisfies

$$\Pr\left[\mathsf{Exp}^{\overline{\text{EUF-ID-CMA}}}_{\mathsf{IBS},\mathcal{A}}(\kappa) \Rightarrow 1\right] \leq \mathsf{negl}(\kappa),$$

where the security experiment $\mathsf{Exp}^{\overline{\text{EUF-ID-CMA}}}_{\mathsf{IBS},\mathcal{A}}$ is defined in Fig. 3.

EUF-ID-CMA vs. ***$\overline{\text{EUF-ID-CMA}}$***. The main difference between the two definitions is that under EUF-ID-CMA the challenger records all generated signing keys in a list $\mathcal{Q}_{sk}$ and reuses them to answer $\mathcal{O}_{\mathsf{KeyGen}}$ and $\mathcal{O}_{\mathsf{Sign}}$ queries, whereas under $\overline{\text{EUF-ID-CMA}}$ the challenger does not store generated signing keys and instead answers $\mathcal{O}_{\mathsf{KeyGen}}$ and $\mathcal{O}_{\mathsf{Sign}}$ queries by sampling a fresh user secret signing key.

- If algorithm KeyGen is a deterministic one, i.e., the signing key is derived deterministically from the master secret key and the identity, then the two security notions are equivalent.
- If algorithm KeyGen is a randomized one, they are different from each other, as shown in the next subsection.

In practice, a user asks for a signing key from the key generation center, and then he/she will store this signing key and reuse it for signing different messages, rather than impractically asking for a fresh signing key from the key generation center for each signing operation. Therefore, EUF-ID-CMA captures a more realistic setting than $\overline{\text{EUF-ID-CMA}}$.

Remark 1. It is hard to prove that the IBS scheme in [13] achieves the standard EUF-ID-CMA security using the proof strategy from that work. The obstacle is as follows. In the standard EUF-ID-CMA experiment, we consider $\mathcal{A}$ with the following strategy: $\mathcal{A}$ does not query the key generation oracle for sk_{id}, but makes two signing queries (id, m_1) and (id, m_2).

- In the standard EUF-ID-CMA experiment, the signing oracle will use the same signing key $sk_{id} = (r, sk'_{id'})$ with $sk'_{id'} \leftarrow \mathsf{KeyExt}'(msk', id')$, and $id' := \mathsf{CHash}(hk, id; r)$ to answer signing queries. As a result, the signing oracle will reply (id, m_1) with (r, s_1, σ_1) and reply (id, m_2) with (r, s_2, σ_2). Note that the randomness r is the same in both responses.
- However, in the security reduction of [13], the reduction algorithm $\mathcal{B}$ replies to (id, m_1) with (r_1, s_1, σ_1) and replies to (id, m_2) with (r_2, s_2, σ_2). According to Fig. 4 (cf. line 35) in [13], r_1 and r_2 are uniform, due to the definition of chameleon hash function (cf. the last line of Definition 1) in [13]. Therefore, $r_1 \neq r_2$ with overwhelming probability.

It is easy to see that $\mathcal{A}$ can distinguish the above two cases. Therefore, $\mathcal{B}$ cannot simulate the standard EUF-ID-CMA experiment for $\mathcal{A}$ and the security proof fails.

$\underline{(mvk, msk) \leftarrow \mathsf{Setup}(1^\kappa).}$

1. $(vk, sk) \leftarrow \mathsf{Sig}_1.\mathsf{KeyGen}(1^\kappa)$.
2. Return $mvk := vk$ and $msk := (vk, sk)$.

$\underline{sk_\sigma \leftarrow \mathsf{KeyGen}(msk, \sigma).}$

1. $(ovk, osk) \leftarrow \mathsf{Sig}_2.\mathsf{KeyGen}(1^\kappa)$.
2. Compute $\Sigma_1 \leftarrow \mathsf{Sig}_1.\mathsf{Sign}(sk, (ovk, \sigma))$.
3. Return $sk_\sigma := (\sigma, (ovk, osk), \Sigma_1)$.

$\underline{\Sigma \leftarrow \mathsf{Sign}(sk_\sigma, m).}$

1. Parse $sk_\sigma = (\sigma, (ovk, osk), \Sigma_1)$.
2. Compute $\Sigma_2 \leftarrow \mathsf{Sig}_2.\mathsf{Sign}(osk, m)$.
3. Return $\Sigma := (ovk, \Sigma_1, \Sigma_2)$.

$\underline{0/1 \leftarrow \mathsf{Ver}(mvk, \sigma, m, \Sigma).}$

1. Parse $\Sigma = (ovk, \Sigma_1, \Sigma_2)$.
2. Return $\mathsf{Sig}_1.\mathsf{Ver}(vk, (ovk, \sigma), \Sigma_1) \wedge \mathsf{Sig}_2.\mathsf{Ver}(ovk, m, \Sigma_2)$.

Fig. 4. Construction of IBS scheme $\overline{\mathsf{IBS}}$ from two signature schemes Sig_1 and Sig_2.

3.2. Separation between EUF-ID-CMA and $\overline{\text{EUF-ID-CMA}}$

Recall that a generic construction of IBS from two signature schemes was proposed in [3], and this construction produces randomized signing keys (KeyGen is a randomized algorithm).

Based on the construction in [3], we present an IBS scheme $\overline{\mathsf{IBS}}$ and prove its $\overline{\text{EUF-ID-CMA}}$ security. We also show that it is not EUF-ID-CMA secure by showing an attack. In this way, we establish a separation between EUF-ID-CMA and $\overline{\text{EUF-ID-CMA}}$ (cf. the following theorem).

Theorem 2. *If Sig_1 is an EUF-CMA secure signature scheme and Sig_2 is the Lamport one-time signature (cf. Section 2.4), then the IBS scheme $\overline{\mathsf{IBS}}$ in Fig. 4 is $\overline{\text{EUF-ID-CMA}}$ secure but not EUF-ID-CMA secure.*

Proof of Theorem 2. Consider the scheme $\overline{\mathsf{IBS}}$ in Fig. 4, which is constructed from an EUF-CMA signature Sig_1 and the Lamport one-time signature scheme Sig_2 (cf. Section 2.4).

We first prove the $\overline{\text{EUF-ID-CMA}}$ security for $\overline{\mathsf{IBS}}$ via two security reductions. Suppose there exists an adversary $\mathcal{A}$ breaking the $\overline{\text{EUF-ID-CMA}}$ security. We construct a PPT algorithm $\mathcal{B}_1$ breaking the EUF-CMA security of Sig_1 or $\mathcal{B}_2$ breaking the EUF-OT-CMA security of Sig_2. Suppose that $\mathcal{A}$ wins with a valid forgery $(\sigma^*, m^*, \Sigma^* = (ovk^*, \Sigma_1^*, \Sigma_2^*))$ for user σ^*, then

$$(\sigma^*, m^*) \notin \mathcal{Q}_{\mathsf{Sign}} \wedge \mathsf{Sig}_1.\mathsf{Ver}(vk, (ovk^*, \sigma^*), \Sigma_1^*) = 1 \wedge \mathsf{Sig}_2.\mathsf{Ver}(ovk^*, m^*, \Sigma_2^*) = 1. \tag{3.1}$$

Since in the $\overline{\text{EUF-ID-CMA}}$ game the signing oracle generates a fresh key pair (ovk, osk) for each query, ovk^* can appear in at most one oracle response. We consider two cases.

- **Case 1:** $(ovk^*, \cdot, \cdot)$ is not the reply of any signing oracle query. Then (ovk^*, σ^*) was never signed by Sig_1. We construct an algorithm $\mathcal{B}_1$ that outputs (ovk^*, σ^*) as the fresh message and Σ_1^* as the corresponding forged signature for Sig_1. By (3.1), Σ_1^* is a valid Sig_1 signature for message (ovk^*, σ^*). In this way $\mathcal{B}_1$ breaks the EUF-CMA security of Sig_1.
- **Case 2:** $(ovk^*, \cdot, \cdot)$ is the reply of some signing oracle query. Let (σ', m') denote the signing query whose response contains ovk^*. We further consider two subcases.
  - If $\sigma' \neq \sigma^*$, then the only message signed by Sig_1 involving ovk^* is $(ovk^*, \sigma') \neq (ovk^*, \sigma^*)$. Hence (ovk^*, σ^*) was never signed, and we can construct $\mathcal{B}_1$ that outputs (ovk^*, σ^*) as the fresh message and Σ_1^* as the forged signature. In this way $\mathcal{B}_1$ breaks the EUF-CMA security of Sig_1.
  - If $\sigma' = \sigma^*$, then since $(\sigma^*, m^*) \notin \mathcal{Q}_{\mathsf{Sign}}$ and the query that generated ovk^* was (σ^*, m'), it follows that $m^* \neq m'$. Now we can construct an algorithm $\mathcal{B}_2$ that breaks the EUF-OT-CMA security of Sig_2. $\mathcal{B}_2$ proceeds as follows. It receives the challenge verification key ovk' for Sig_2 from its own challenger, runs $\mathsf{Sig}_1.\mathsf{KeyGen}$ to obtain (vk, sk), and chooses uniformly at random an index $i \xleftarrow{\$} [Q_s]$, where Q_s is the number of signing queries made by $\mathcal{A}$. For each signing query except the ith on (σ, m), $\mathcal{B}_2$ simulates the signing oracle by generating a fresh one-time key pair (ovk, osk), computing $\Sigma_1 \leftarrow \mathsf{Sig}_1.\mathsf{Sign}(sk, (ovk, \sigma))$, obtaining $\Sigma_2 \leftarrow \mathsf{Sig}_2.\mathsf{Sign}(osk, m)$, and returning $\Sigma = (ovk, \Sigma_1, \Sigma_2)$ to $\mathcal{A}$. For the ith signing query on (σ', m'), $\mathcal{B}_2$ computes $\Sigma_1' \leftarrow \mathsf{Sig}_1.\mathsf{Sign}(sk, (ovk', \sigma'))$, forwards m' to its own Sig_2-signing oracle, receives Σ_2', and returns $\Sigma' = (ovk', \Sigma_1', \Sigma_2')$ (using the challenge ovk' in place of a freshly generated ovk). When $\mathcal{A}$ outputs a forgery $(\sigma^*, m^*, \Sigma^* = (ovk^*, \Sigma_1^*, \Sigma_2^*))$, $\mathcal{B}_2$ checks whether $ovk^* = ovk'$. If $ovk^* \neq ovk'$, $\mathcal{B}_2$ aborts. Since i is

chosen uniformly, $ovk^* = ovk'$ holds with probability $1/Q_s$. In case of non-abort, $\mathcal{B}_2$ outputs (m^*, Σ_2^*) as its own forgery. Since $m^* \neq m'$ and (3.1) implies (m^*, Σ_2^*) is a valid Sig_2 forgery under ovk', $\mathcal{B}_2$ breaks the EUF-OT-CMA security of Sig_2.

However, $\overline{\mathsf{IBS}}$ does not achieve EUF-ID-CMA security. An adversary can break the EUF-ID-CMA security as follows. For the same identity σ, the adversary queries the signing oracle on two distinct messages m and m' (cf. Fact 1), and receives signatures $\Sigma = (ovk, \Sigma_1, \Sigma_2)$ and $\Sigma' = (ovk, \Sigma_1, \Sigma_2')$, since the challenger reuses the same secret signing key for σ. Using these two signatures, the adversary can produce a new one-time signature Σ_2^* on $m^* \notin \{m, m'\}$ for identity σ. Hence $\Sigma^* = (ovk, \Sigma_1, \Sigma_2^*)$ is a valid forgery for $\overline{\mathsf{IBS}}$, showing that the scheme is not EUF-ID-CMA-secure. □

According to the last part of the proof, we have the following fact.

Fact 2. $\overline{\mathsf{IBS}}$ does not achieve EUF-ID-CMA security. Given two distinct messages m and m' and their corresponding signatures Σ and Σ' under the same identity, the adversary can forge a valid signature on a new message $m^* \notin \{m, m'\}$.

3.3. EUF-ID-CMA secure IBS from lattices in the standard model

In this subsection, we provide an IBS scheme from lattices. We also prove the EUF-ID-CMA security for our IBS scheme based on the SIS assumption. Let $n, q, \hat{x}, \hat{y}, \hat{t}, \tilde{x}, \tilde{y}, \tilde{t}, \ell$ be positive integers, and let $w = n\lceil \log q \rceil$. The following building blocks are used in our IBS construction.

- A PRF $\widehat{\mathsf{PRF}} = (\widehat{\mathsf{PRF}}.\mathsf{Setup}, \widehat{\mathsf{PRF}})$ with key space $\{0,1\}^{\hat{t}}$, input space $\{0,1\}^{\hat{x}}$ and output space $\{0,1\}^{\hat{y}}$.
- A PRF $\widetilde{\mathsf{PRF}} = (\widetilde{\mathsf{PRF}}.\mathsf{Setup}, \widetilde{\mathsf{PRF}})$ with key space $\{0,1\}^{\tilde{t}}$, input space $\{0,1\}^{\tilde{x}}$ and output space $\{0,1\}^{\tilde{y}}$, where $\tilde{x} = \hat{x} + \ell + \kappa$.
- A matching function $f_{\mathbf{r}} : \{0,1\}^{|\mathbf{r}|} \to \{0,1\}$ specified by some vector $\mathbf{r} \in \{0,1\}^{|\mathbf{r}|}$, s.t., for any $\mathbf{m} \in \{0,1\}^{|\mathbf{r}|}$:

$$f_{\mathbf{r}}(\mathbf{m}) = 1 \text{ if } \mathbf{r} = \mathbf{m} \text{ and } f_{\mathbf{r}}(\mathbf{m}) = 0 \text{ otherwise.} \tag{3.2}$$

Our IBS construction is given in Fig. 5.

Here we give a more detailed description of our parameter setting. Parameters of our IBS include the dimension parameters $\hat{x}, \hat{y}, \hat{t}, \tilde{x}, \tilde{y}, \tilde{t}, \ell$, the SIS parameters n, m, q, β and Gaussian parameters δ, γ. We require that parameters satisfy the following conditions simultaneously.

- Let $\hat{x}, \hat{y}, \hat{t}, \tilde{x}, \tilde{y}, \tilde{t}, \ell = \mathsf{poly}(\kappa)$ be positive integers and $\tilde{x} = \hat{x} + \ell + \kappa$.
- Let n, m, q, β be positive parameters, $m, \beta = \mathsf{poly}(\kappa)$, $\beta \cdot \omega(\sqrt{n \log n}) \leq q$, so that the SIS problem is hard according to Lemma 8 (the SIS assumption).
- Let $m = O(n \log q)$, $\gamma \geq O(\kappa^c) \cdot \omega(\sqrt{\log n})$ for some constant c and $\gamma \geq O(\sqrt{n \log q})$ so that Lemma 1 (trapdoor generation) and Lemma 4 (basis randomization) can be applied.
- Let $\delta \geq O(\kappa^c) \cdot \omega(\sqrt{\log m})$ for some constant c and $\delta \geq O(n \log q) \cdot \omega(\sqrt{\log m})$ so that Lemma 2 (preimage sampling) can be applied.
- Let $\beta \geq O(\kappa^c) \cdot \delta\sqrt{m + 2w}$ for some constant c, which is required for our proof of Theorem 3.

The correctness of IBS follows directly from Lemma 1 (trapdoor generation), Lemma 2 (preimage sampling) and Lemma 7 (homomorphic evaluation).

Theorem 3. *Let $\widehat{\mathsf{PRF}}$ and $\widetilde{\mathsf{PRF}}$ be secure pseudo-random functions. Given the parameters described above, scheme* IBS *in Fig. 5 has EUF-ID-CMA security if the $\mathsf{SIS}_{n,q,\beta,m}$ assumption holds. More precisely, the EUF-ID-CMA security of* IBS *is reduced to the SIS assumption and the pseudo-randomness of $\widehat{\mathsf{PRF}}$ and $\widetilde{\mathsf{PRF}}$:*

$$\mathsf{Adv}^{\textit{EUF-ID-CMA}}_{\mathsf{IBS},\mathcal{A}}(\kappa) \leq Q_s \mathsf{Adv}^{\mathsf{SIS}}_{[n,q,\beta,m]}(\kappa) + 2\mathsf{Adv}^{pse}_{\widehat{\mathsf{PRF}}}(\kappa) + 2\mathsf{Adv}^{pse}_{\widetilde{\mathsf{PRF}}}(\kappa) + \mathsf{negl}(\kappa),$$

where Q_s is the number of signing queries issued by $\mathcal{A}$.

Proof of Theorem 3. The proof is accomplished with hybrid games G_0-G_7, where G_0 is identical to $\mathsf{Exp}^{\text{EUF-ID-CMA}}_{\mathsf{IBS},\mathcal{A}}(\kappa)$ (cf. Fig. 3) from the view of adversary $\mathcal{A}$. Let $\Pr[\mathsf{G}_i \Rightarrow 1]$ denote the probability that game G_i outputs 1. We aim to prove that

$$\Pr[\mathsf{Exp}^{\text{EUF-ID-CMA}}_{\mathsf{IBS},\mathcal{A}}(\kappa) \Rightarrow 1] = \Pr[\mathsf{G}_0 \Rightarrow 1] = \mathsf{negl}(\kappa).$$

To do this, we will show that games G_i and G_{i-1} are indistinguishable for all $i \in [7] \setminus \{4\}$; that $\Pr[\mathsf{G}_4 \Rightarrow 1] = \frac{1}{Q_s+1} \cdot \Pr[\mathsf{G}_3 \Rightarrow 1]$; and that $\Pr[\mathsf{G}_7 \Rightarrow 1] = \mathsf{negl}(\kappa)$. Next, we will present and analyze these games. The differences between adjacent games are highlighted with gray shading.

Game G_0. Game G_0 is identical to $\mathsf{Exp}^{\text{EUF-ID-CMA}}_{\mathsf{IBS},\mathcal{A}}(\kappa)$ defined in Fig. 3 except that the challenger additionally samples $i^* \xleftarrow{\$} [Q_s + 1]$, and samples $\hat{\mathbf{k}} \xleftarrow{\$} \{0,1\}^{\hat{t}}$, $\tilde{\mathbf{k}} \xleftarrow{\$} \{0,1\}^{\tilde{t}}$ as secret keys for $\widehat{\mathsf{PRF}}$ and $\widetilde{\mathsf{PRF}}$ respectively. Meanwhile, the challenger introduces a counter cnt to index its signing queries.

0. Challenger $\mathcal{C}$ initializes sets $Q_{\mathsf{KeyGen}} := \emptyset$, $Q_{\mathsf{Sign}} := \emptyset$, $Q_{sk} := \emptyset$, samples $i^* \xleftarrow{\$} [Q_s + 1]$, samples $\hat{\mathbf{k}} \xleftarrow{\$} \{0,1\}^{\hat{t}}, \tilde{\mathbf{k}} \xleftarrow{\$} \{0,1\}^{\tilde{t}}$ as secret keys for $\widehat{\mathsf{PRF}}$ and $\widetilde{\mathsf{PRF}}$ and initializes the counter $\mathsf{cnt} := 0$.

$(mvk, msk) \leftarrow \mathsf{Setup}(1^\kappa)$.

1. Generate $\hat{pp} \leftarrow \widehat{\mathsf{PRF}}.\mathsf{Setup}(1^\kappa)$ and $\tilde{pp} \leftarrow \widetilde{\mathsf{PRF}}.\mathsf{Setup}(1^\kappa)$.
2. Generate $(\mathbf{A}, \mathbf{T}_\mathbf{A}) \leftarrow \mathsf{TrapGen}(1^n, 1^m, q)$ with $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ and $\mathbf{T}_\mathbf{A} \in \mathbb{Z}^{m\times m}$.
3. For $i \in [\hat{t}]$, sample $\hat{\mathbf{A}}_i \xleftarrow{\$} \mathbb{Z}_q^{n\times w}$. For $i \in [\tilde{t}]$, sample $\tilde{\mathbf{A}}_i \xleftarrow{\$} \mathbb{Z}_q^{n\times w}$.
4. Set $mvk := (\mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]}, \hat{pp}, \tilde{pp})$, $msk := (mvk, \mathbf{T}_\mathbf{A})$.
5. Return (mvk, msk).

$sk_\sigma \leftarrow \mathsf{KeyGen}(msk, \sigma \in \{0,1\}^{\hat{x}})$.

1. Parse $msk = (mvk, \mathbf{T}_\mathbf{A})$ and $mvk = (\mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]}, \hat{pp}, \tilde{pp})$.
2. Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]})$.
3. Sample $\hat{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\hat{y}}$. Construct $f_{\hat{\mathbf{r}}}$ as defined by (3.2).
4. Compute $\mathbf{C}_{f_{\hat{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i\in[\hat{y}]})$.
5. Delegate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}], \mathbf{T}_\mathbf{A})$.
6. Randomize $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{RandBasis}(\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \gamma)$.
7. Return $sk_\sigma := (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}}, \mathbf{C}_{f_{\hat{\mathbf{r}}}}, \sigma)$.

$\Sigma \leftarrow \mathsf{Sign}(sk_\sigma, \mathbf{m} \in \{0,1\}^\ell)$.

1. Sample $\mathbf{z} \xleftarrow{\$} \{0,1\}^\kappa$ and set $\mathbf{m}' := \mathbf{m}\|\mathbf{z}$. Parse $sk_\sigma = (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}}, \mathbf{C}_{f_{\hat{\mathbf{r}}}}, \sigma)$.
2. Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widetilde{\mathsf{PRF}}(\cdot, \sigma\|\mathbf{m}'), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]})$.
3. Sample $\tilde{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\tilde{y}}$. Construct $f_{\tilde{\mathbf{r}}}$ as defined by (3.2).
4. Compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i\in[\tilde{y}]})$.
5. Delegate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}})$.
6. Sample $\mathbf{s} \leftarrow \mathsf{SamplePre}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}, \mathbf{0}^n, \delta)$ with $\mathbf{s} \in \mathbb{Z}_q^{m+2w}$ and $\mathbf{s} \neq \mathbf{0}^{m+2w}$.
7. Return $\Sigma := (\mathbf{s}, \hat{\mathbf{r}}, \tilde{\mathbf{r}}, \mathbf{z})$.

$0/1 \leftarrow \mathsf{Ver}(mvk, \sigma, \mathbf{m}, \Sigma)$.

1. Parse $\Sigma = (\mathbf{s}, \hat{\mathbf{r}}, \tilde{\mathbf{r}}, \mathbf{z})$. Set $\mathbf{m}' := \mathbf{m}\|\mathbf{z}$.
2. Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]})$.
3. Construct $f_{\hat{\mathbf{r}}}$ as defined by (3.2). Compute $\mathbf{C}_{f_{\hat{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i\in[\hat{y}]})$.
4. Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widetilde{\mathsf{PRF}}(\cdot, \sigma\|\mathbf{m}'), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]})$.
5. Construct $f_{\tilde{\mathbf{r}}}$ as defined by (3.2). Compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i\in[\tilde{y}]})$.
6. If $[\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}] \cdot \mathbf{s} = \mathbf{0}^n$ and $0 < \|\mathbf{s}\| \leq \delta\sqrt{m+2w}$, return 1; otherwise, return 0.

Fig. 5. The lattice-based IBS construction.

1. During the setup phase, challenger $\mathcal{C}$ proceeds as follows.
   - Generate $\hat{pp} \leftarrow \widehat{\mathsf{PRF}}.\mathsf{Setup}(1^\kappa)$ and $\tilde{pp} \leftarrow \widetilde{\mathsf{PRF}}.\mathsf{Setup}(1^\kappa)$.
   - Generate $(\mathbf{A}, \mathbf{T}_\mathbf{A}) \leftarrow \mathsf{TrapGen}(1^n, 1^m, q)$.
   - Sample $\hat{\mathbf{A}}_i \xleftarrow{\$} \mathbb{Z}_q^{n\times w}$ for $i \in [\hat{t}]$ and $\tilde{\mathbf{A}}_i \xleftarrow{\$} \mathbb{Z}_q^{n\times w}$ for $i \in [\tilde{t}]$.
   - Set $mvk = (\mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]}, \hat{pp}, \tilde{pp})$ and $msk = (mvk, \mathbf{T}_\mathbf{A}, \hat{\mathbf{k}}, \tilde{\mathbf{k}})$ and return mvk to $\mathcal{A}$.
2. On receiving a key generation query σ from $\mathcal{A}$, $\mathcal{C}$ proceeds as follows.
   - Add σ to $\mathcal{Q}_{\mathsf{KeyGen}}$. If $\exists(sk_\sigma, \sigma) \in \mathcal{Q}_{sk}$, return sk_σ.
   - Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]})$.
   - Sample $\hat{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\hat{y}}$ and construct $f_{\hat{\mathbf{r}}}$ as defined by (3.2).
   - Compute $\mathbf{C}_{f_{\hat{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i\in[\hat{y}]})$.
   - Delegate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}], \mathbf{T}_\mathbf{A})$ and randomize it by $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{RandBasis}(\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \gamma)$.
   - Return $sk_\sigma = (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}}, \mathbf{C}_{f_{\hat{\mathbf{r}}}}, \sigma)$ to $\mathcal{A}$ and add (sk_σ, σ) to $\mathcal{Q}_{sk}$.
3. On receiving a signing query $(\sigma, \mathbf{m})$ from $\mathcal{A}$, $\mathcal{C}$ proceeds as follows.
   - Add $(\sigma, \mathbf{m})$ to $\mathcal{Q}_{\mathsf{Sign}}$ and update $\mathsf{cnt} := \mathsf{cnt} + 1$.
   - Sample $\mathbf{z} \xleftarrow{\$} \{0,1\}^\kappa$ and set $\mathbf{m}' := \mathbf{m}\|\mathbf{z}$.
   - If $\nexists(sk_\sigma, \sigma) \in \mathcal{Q}_{sk}$, then generate sk_σ by invoking the key generation query on σ as in $\mathcal{O}_{\mathsf{KeyGen}}$.
   - Parse $sk_\sigma = (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}}, \mathbf{C}_{f_{\hat{\mathbf{r}}}}, \sigma)$.
   - Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widetilde{\mathsf{PRF}}(\cdot, \sigma\|\mathbf{m}'), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]})$.

- Sample $\tilde{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\tilde{y}}$. Construct $f_{\tilde{\mathbf{r}}}$ as defined by (3.2).
- Compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i\in[\tilde{y}]})$.
- Delegate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}})$.
- Sample $\mathbf{s} \leftarrow \mathsf{SamplePre}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}, \mathbf{0}^n, \delta)$ with $\mathbf{s} \in \mathbb{Z}_q^{m+2w}$ and $\mathbf{s} \neq \mathbf{0}^{m+2w}$.
- Return $\Sigma = (\mathbf{s}, \hat{\mathbf{r}}, \tilde{\mathbf{r}}, \mathbf{z})$.

4. On receiving a forgery $(\sigma^*, \mathbf{m}^*, \Sigma^*)$ from $\mathcal{A}$, $\mathcal{C}$ proceeds as follows.
   - If $\sigma^* \in \mathcal{Q}_{\mathsf{KeyGen}}$ or $(\sigma^*, \mathbf{m}^*) \in \mathcal{Q}_{\mathsf{Sign}}$, G_0 outputs 0.
   - Parse $\Sigma^* = (\mathbf{s}^*, \hat{\mathbf{r}}^*, \tilde{\mathbf{r}}^*, \mathbf{z}^*)$ and set $\mathbf{m}^{*\prime} := \mathbf{m}^* \| \mathbf{z}^*$.
   - Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma^*), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{l}]})$.
   - Construct $f_{\hat{\mathbf{r}}^*}$ and compute $\mathbf{C}_{f_{\hat{\mathbf{r}}^*}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}^*}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i\in[\hat{y}]})$.
   - Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widetilde{\mathsf{PRF}}(\cdot, \sigma^* \| \mathbf{m}^{*\prime}), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{l}]})$.
   - Construct $f_{\tilde{\mathbf{r}}^*}$ and compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}^*}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i\in[\tilde{y}]})$.
   - If $[\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}^*}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^* = \mathbf{0}^n$ and $0 < \|\mathbf{s}^*\| \leq \delta\sqrt{m+2w}$, G_0 outputs 1; otherwise, G_0 outputs 0.

Game G_1**.** Game G_1 is identical to G_0 except for the generation of $\hat{\mathbf{r}}$ in the key generation oracle $\mathcal{O}_{\mathsf{KeyGen}}$ and $\tilde{\mathbf{r}}$ in the signing oracle $\mathcal{O}_{\mathsf{Sign}}$. More precisely, in G_1, the challenger computes $\tilde{\mathbf{r}} := \widetilde{\mathsf{PRF}}(\tilde{\mathbf{k}}, \sigma_{i^*} \| \mathbf{m}')$ when replying to the i^*th $\mathcal{O}_{\mathsf{Sign}}(\sigma_{i^*}, \cdot)$ query and the later $\mathcal{O}_{\mathsf{Sign}}(\sigma = \sigma_{i^*}, \cdot)$ queries, rather than sampling $\tilde{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\tilde{y}}$ as in G_0. Meanwhile, for the later $\mathcal{O}_{\mathsf{KeyGen}}(\sigma \neq \sigma_{i^*})$ queries, the challenger computes $\hat{\mathbf{r}} := \widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma)$ for the reply, rather than sampling $\hat{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\hat{y}}$ as in G_0.

0′. Challenger $\mathcal{C}$ initializes sets $\mathcal{Q}_{\mathsf{KeyGen}} := \emptyset$, $\mathcal{Q}_{\mathsf{Sign}} := \emptyset$, $\mathcal{Q}_{sk} := \emptyset$, samples $i^* \xleftarrow{\$} [Q_s + 1]$, samples $\hat{\mathbf{k}} \xleftarrow{\$} \{0,1\}^{\hat{l}}$, $\tilde{\mathbf{k}} \xleftarrow{\$} \{0,1\}^{\tilde{l}}$ as secret keys for $\widehat{\mathsf{PRF}}$ and $\widetilde{\mathsf{PRF}}$ and initializes the counter $\mathsf{cnt} := 0$ and identity $\sigma_{i^*} := \perp$.

2′. On receiving a key generation query σ from $\mathcal{A}$, $\mathcal{C}$ proceeds as follows.
   - Add σ to $\mathcal{Q}_{\mathsf{KeyGen}}$. If $\exists (sk_\sigma, \sigma) \in \mathcal{Q}_{sk}$, return sk_σ.
   - Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{l}]})$.
   - If $\sigma \neq \sigma_{i^*}$, compute $\hat{\mathbf{r}} := \widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma)$; else sample $\hat{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\hat{y}}$ and construct $f_{\hat{\mathbf{r}}}$ as defined by (3.2).
   - Compute $\mathbf{C}_{f_{\hat{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i\in[\hat{y}]})$.
   - Delegate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}})$ and randomize it by $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{RandBasis}(\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \gamma)$.
   - Return $sk_\sigma = (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}}, \mathbf{C}_{f_{\hat{\mathbf{r}}}}, \sigma)$ to $\mathcal{A}$ and add (sk_σ, σ) to $\mathcal{Q}_{sk}$.

3′. On receiving a signing query $(\sigma, \mathbf{m})$ from $\mathcal{A}$, $\mathcal{C}$ proceeds as follows.
   - Add $(\sigma, \mathbf{m})$ to $\mathcal{Q}_{\mathsf{Sign}}$ and update $\mathsf{cnt} := \mathsf{cnt} + 1$.
   - Sample $\mathbf{z} \xleftarrow{\$} \{0,1\}^{\kappa}$ and set $\mathbf{m}' := \mathbf{m} \| \mathbf{z}$.
   - If $\mathsf{cnt} = i^*$, set $\sigma_{i^*} := \sigma$.
   - If $\nexists (sk_\sigma, \sigma) \in \mathcal{Q}_{sk}$, then generate sk_σ by invoking the key generation query on σ as in $\mathcal{O}_{\mathsf{KeyGen}}$.
   - Parse $sk_\sigma = (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}}, \mathbf{C}_{f_{\hat{\mathbf{r}}}}, \sigma)$.
   - Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widetilde{\mathsf{PRF}}(\cdot, \sigma \| \mathbf{m}'), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{l}]})$.
   - If $\sigma = \sigma_{i^*}$, compute $\tilde{\mathbf{r}} \leftarrow \widetilde{\mathsf{PRF}}(\tilde{\mathbf{k}}, \sigma \| \mathbf{m}')$; else sample $\tilde{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\tilde{y}}$. Construct $f_{\tilde{\mathbf{r}}}$ as defined by (3.2).
   - Compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i\in[\tilde{y}]})$.
   - Delegate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}})$.
   - Sample $\mathbf{s} \leftarrow \mathsf{SamplePre}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}, \mathbf{0}^n, \delta)$ with $\mathbf{s} \in \mathbb{Z}_q^{m+2w}$ and $\mathbf{s} \neq \mathbf{0}^{m+2w}$.
   - Return $\Sigma = (\mathbf{s}, \hat{\mathbf{r}}, \tilde{\mathbf{r}}, \mathbf{z})$.

Lemma 11. $|\Pr[\mathsf{G}_1(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_0(\kappa) \Rightarrow 1]| \leq \mathsf{Adv}^{pse}_{\widehat{\mathsf{PRF}}}(\kappa) + \mathsf{Adv}^{pse}_{\widetilde{\mathsf{PRF}}}(\kappa)$.

Proof of Lemma 11. According to the pseudo-randomness of $\widehat{\mathsf{PRF}}$ and $\widetilde{\mathsf{PRF}}$, the distributions of $\hat{\mathbf{r}} := \widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma)$ and $\tilde{\mathbf{r}} := \widetilde{\mathsf{PRF}}(\tilde{\mathbf{k}}, \sigma \| \mathbf{m}')$ in G_1 are computationally indistinguishable from $\hat{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\hat{y}}$ and $\tilde{\mathbf{r}} \xleftarrow{\$} \{0,1\}^{\tilde{y}}$ in G_0. Thus, Lemma 11 holds. □

Game G_2**.** G_2 is identical to G_1 except that challenger $\mathcal{C}$ adds a forgery-rejection rule upon receiving a forgery $(\sigma^*, \mathbf{m}^*, \Sigma^*)$ from $\mathcal{A}$.

4′. On receiving a forgery $(\sigma^*, \mathbf{m}^*, \Sigma^*)$ from $\mathcal{A}$, $\mathcal{C}$ proceeds as follows.
   - If $\sigma^* \in \mathcal{Q}_{\mathsf{KeyGen}}$ or $(\sigma^*, \mathbf{m}^*) \in \mathcal{Q}_{\mathsf{Sign}}$, G_2 outputs 0.
   - Parse $\Sigma^* = (\mathbf{s}^*, \hat{\mathbf{r}}^*, \tilde{\mathbf{r}}^*, \mathbf{z}^*)$ and set $\mathbf{m}^{*\prime} := \mathbf{m}^* \| \mathbf{z}^*$.
   - **Forgery-rejection rule 1**: Compute $\hat{\mathbf{r}} := \widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma^*)$ and $\tilde{\mathbf{r}} := \widetilde{\mathsf{PRF}}(\tilde{\mathbf{k}}, \sigma^* \| \mathbf{m}^{*\prime})$. If $\hat{\mathbf{r}}^* = \hat{\mathbf{r}}$ or $\tilde{\mathbf{r}}^* = \tilde{\mathbf{r}}$, G_2 outputs 0.
   - Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma^*), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{l}]})$.
   - Construct $f_{\hat{\mathbf{r}}^*}$ and compute $\mathbf{C}_{f_{\hat{\mathbf{r}}^*}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}^*}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i\in[\hat{y}]})$.
   - Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widetilde{\mathsf{PRF}}(\cdot, \sigma^* \| \mathbf{m}^{*\prime}), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{l}]})$.

– Construct $f_{\tilde{\mathbf{r}}^*}$ and compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}^*}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i\in[\tilde{y}]})$.
– If $[\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}}|\mathbf{C}_{f_{\hat{\mathbf{r}}^*}}] \cdot \mathbf{s}^* = \mathbf{0}^n$ and $0 < \|\mathbf{s}^*\| \le \delta\sqrt{m+2w}$, G_2 outputs 1; otherwise, G_2 outputs 0.

Lemma 12. $|\Pr[\mathsf{G}_2(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_1(\kappa) \Rightarrow 1]| \le \mathsf{Adv}^{pse}_{\widehat{\mathsf{PRF}}}(\kappa) + \mathsf{Adv}^{pse}_{\widetilde{\mathsf{PRF}}}(\kappa) + \mathsf{negl}(\kappa)$.

Proof of Lemma 12. If $\mathcal{A}$ wins (i.e., the game outputs 1), then $\sigma^* \notin \mathcal{Q}_{\mathsf{KeyGen}}$ and $(\sigma^*, \mathbf{m}^*) \notin \mathcal{Q}_{\mathsf{Sign}}$, which implies $\mathcal{A}$ never queries the key generation oracle $\mathcal{O}_{\mathsf{KeyGen}}$ on input σ^* and never queries the signing oracle $\mathcal{O}_{\mathsf{Sign}}$ on the pair $(\sigma^*, \mathbf{m}^*)$. Then Lemma 12 follows from the pseudo-randomness of $\widehat{\mathsf{PRF}}$ and $\widetilde{\mathsf{PRF}}$.

More precisely, define event Bad as $\hat{\mathbf{r}}^* = \hat{\mathbf{r}} \vee \tilde{\mathbf{r}}^* = \tilde{\mathbf{r}}$. Note that G_2 is identical to G_1 when event Bad does not occur, so

$$|\Pr[\mathsf{G}_2(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_1(\kappa) \Rightarrow 1]| \le \Pr[\mathsf{Bad}] \le \Pr[\hat{\mathbf{r}} = \hat{\mathbf{r}}^*] + \Pr[\tilde{\mathbf{r}} = \tilde{\mathbf{r}}^*]$$

It remains to bound $\Pr[\hat{\mathbf{r}}^* = \hat{\mathbf{r}}]$ and $\Pr[\tilde{\mathbf{r}}^* = \tilde{\mathbf{r}}]$. Given an adversary $\mathcal{A}$ such that $\hat{\mathbf{r}}^* = \hat{\mathbf{r}}$, we construct an algorithm $\mathcal{B}$ that breaks the pseudo-randomness of $\widehat{\mathsf{PRF}}$. $\mathcal{B}$ receives $\hat{pp}$ from its own challenger and has access to oracle $\mathcal{O}_{\mathsf{PRF}}(\cdot)$ and $\mathcal{O}_{\mathsf{Chl}}(\cdot)$. It simulates G_2 for $\mathcal{A}$ as follows.

– In setup, $\mathcal{B}$ uses $\hat{pp}$ supplied by its challenger for mvk. Except for $\hat{pp}$ and $\hat{\mathbf{k}}$, $\mathcal{B}$ simulates the other parts of mvk just like G_2. In this way, $\mathcal{B}$ obtains $mvk = (\mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]}, \hat{pp}, \tilde{pp})$ and $msk = (mvk, \mathbf{T}_{\mathbf{A}}, \cdot, \tilde{\mathbf{k}})$. It sends mvk to $\mathcal{A}$.
– To answer a key-generation query σ (when $\sigma \neq \sigma_{i^*}$), $\mathcal{B}$ queries $\mathcal{O}_{\mathsf{PRF}}(\sigma)$ and uses the returned value as $\hat{\mathbf{r}}$, instead of computing $\hat{\mathbf{r}} := \widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma)$. With the help of msk, $\mathcal{B}$ perfectly simulates the replies for the key generation queries.
– To answer a signing query, $\mathcal{B}$ uses $\tilde{\mathbf{k}}$ and $\mathbf{T}_{\mathbf{A}}$ to perfectly simulate the replies for the signing queries.
– When $\mathcal{A}$ outputs a forgery $(\sigma^*, \mathbf{m}^*, \Sigma^* = (\mathbf{s}^*, \hat{\mathbf{r}}^*, \tilde{\mathbf{r}}^*, \mathbf{z}^*))$, $\mathcal{B}$ queries its own oracle $\mathcal{O}_{\mathsf{Chl}}(\sigma^*)$ and obtains $\hat{\mathbf{r}}$. It then compares $\hat{\mathbf{r}}^*$ with $\hat{\mathbf{r}}$. If $\hat{\mathbf{r}}^* = \hat{\mathbf{r}}$, $\mathcal{B}$ outputs $b' = 1$; otherwise it outputs $b' = 0$.

Note that $\hat{\mathbf{r}}^* = \hat{\mathbf{r}}$ if and only if $\mathcal{B}$ outputs $b' = 1$. If $\mathcal{B}$'s challenger chooses $b = 1$ (with probability 1/2), then $\hat{\mathbf{r}} = \mathsf{PRF}(\hat{\mathbf{k}}, \sigma^*)$ and $\hat{\mathbf{r}}^* = \hat{\mathbf{r}}$ leading to $b' = 1$; If $\mathcal{B}$'s challenger chooses $b = 0$ (with probability 1/2), then $\hat{\mathbf{r}} = \mathsf{RF}(\sigma^*)$ is independently and uniformly chosen and $\hat{\mathbf{r}}^* = \hat{\mathbf{r}}$ (i.e., $b' = 1$) hardly holds. Thus

$$\begin{aligned}
\Pr[\hat{\mathbf{r}}^* = \hat{\mathbf{r}}] &= \Pr[b' = 1] = \Pr[b' = 1 \wedge b = 1] + \Pr[b' = 1 \wedge b = 0] \\
&= \frac{1}{2}\Pr[b' = 1 | b = 1] + \frac{1}{2}\Pr[b' = 1 | b = 0] \\
&\le \frac{1}{2}\Big|\Pr[b' = 1 | b = 1] - \Pr[b' = 1 | b = 0]\Big| + \Pr[b' = 1 | b = 0] \\
&\le \mathsf{Adv}^{pse}_{\widehat{\mathsf{PRF}}}(\kappa) + \mathsf{negl}(\kappa).
\end{aligned}$$

The same reduction applied to $\widetilde{\mathsf{PRF}}$ yields $\Pr[\tilde{\mathbf{r}}^* = \tilde{\mathbf{r}}] \le \mathsf{Adv}^{pse}_{\widetilde{\mathsf{PRF}}}(\kappa) + \mathsf{negl}(\kappa)$.

Combining these bounds gives the claimed inequality and completes the proof. □

Game G_3. Game G_3 is identical to G_2 except for the generations of $\{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]}$ and $\{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]}$. In G_3, $\hat{\mathbf{A}}_i = \mathbf{A}\hat{\mathbf{R}}_i + \hat{k}_i\mathbf{G}$ instead of $\hat{\mathbf{A}}_i \xleftarrow{\$} \mathbb{Z}_q^{n\times w}$ and $\tilde{\mathbf{A}}_i = \mathbf{A}\tilde{\mathbf{R}}_i + \tilde{k}_i\mathbf{G}$ instead of $\tilde{\mathbf{A}}_i \xleftarrow{\$} \mathbb{Z}_q^{n\times w}$.

1′. During the setup phase, challenger C proceeds as follows.
– Generate $\hat{pp} \leftarrow \widehat{\mathsf{PRF}}.\mathsf{Setup}(1^\kappa)$ and $\tilde{pp} \leftarrow \widetilde{\mathsf{PRF}}.\mathsf{Setup}(1^\kappa)$.
– Generate $(\mathbf{A}, \mathbf{T}_{\mathbf{A}}) \leftarrow \mathsf{TrapGen}(1^n, 1^m, q)$;
– Sample $\hat{\mathbf{R}}_i \xleftarrow{\$} \{1,-1\}^{m\times w}$ and set $\hat{\mathbf{A}}_i := \mathbf{A}\hat{\mathbf{R}}_i + \hat{k}_i\mathbf{G}$ for $i \in [\hat{t}]$, sample $\tilde{\mathbf{R}}_i \xleftarrow{\$} \{1,-1\}^{m\times w}$ and set $\tilde{\mathbf{A}}_i := \mathbf{A}\tilde{\mathbf{R}}_i + \tilde{k}_i\mathbf{G}$ for $i \in [\tilde{t}]$.
– Set $mvk = (\mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]}, \hat{pp}, \tilde{pp})$ and $msk = (mvk, \mathbf{T}_{\mathbf{A}}, \hat{\mathbf{k}}, \tilde{\mathbf{k}})$ and return mvk to $\mathcal{A}$.

Lemma 13. $|\Pr[\mathsf{G}_3(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_2(\kappa) \Rightarrow 1]| \le (\hat{t} + \tilde{t}) \cdot \mathsf{negl}(\kappa) = \mathsf{negl}(\kappa)$.

Proof of Lemma 13. According to Lemma 6, we know that

$$\mathsf{SD}(\mathbf{A}\hat{\mathbf{R}}_i + \hat{k}_i\mathbf{G}, \$) \le \mathsf{negl}(\kappa), \mathsf{SD}(\$, \mathbf{A}\hat{\mathbf{R}}_i) \le \mathsf{negl}(\kappa), \mathsf{SD}(\mathbf{A}\tilde{\mathbf{R}}_j + \tilde{k}_j\mathbf{G}, \$) \le \mathsf{negl}(\kappa), \mathsf{SD}(\$, \mathbf{A}\tilde{\mathbf{R}}_j) \le \mathsf{negl}(\kappa)$$

for $\hat{\mathbf{R}}_i, \tilde{\mathbf{R}}_j \xleftarrow{\$} \{1,-1\}^{m\times w}$. Therefore

$$\mathsf{SD}(\mathbf{A}\hat{\mathbf{R}}_i + \hat{k}_i\mathbf{G}, \mathbf{A}\hat{\mathbf{R}}_i) \le \mathsf{negl}(\kappa), \quad \mathsf{SD}(\mathbf{A}\tilde{\mathbf{R}}_j + \tilde{k}_j\mathbf{G}, \mathbf{A}\tilde{\mathbf{R}}_j) \le \mathsf{negl}(\kappa)$$

for $i \in [\hat{t}]$ and $j \in [\tilde{t}]$, which means

$$\mathsf{SD}\Big((\{\mathbf{A}\hat{\mathbf{R}}_i + \hat{k}_i\mathbf{G}\}_{i\in[\hat{t}]}, \{\mathbf{A}\tilde{\mathbf{R}}_j + \tilde{k}_j\mathbf{G}\}_{j\in[\tilde{t}]}), (\{\mathbf{A}\hat{\mathbf{R}}_i\}_{i\in[\hat{t}]}, \{\mathbf{A}\tilde{\mathbf{R}}_j\}_{j\in[\tilde{t}]})\Big) \le (\hat{t} + \tilde{t}) \cdot \mathsf{negl}(\kappa) = \mathsf{negl}(\kappa).$$

Therefore, Lemma 13 follows. □

Game G_4. G_4 is identical to G_3 except that the challenger adds a second forgery-rejection rule upon receiving a forgery $(\sigma^*, \mathbf{m}^*, \Sigma^*)$ from $\mathcal{A}$.

Given $\mathcal{A}$'s forgery $(\sigma^*, \mathbf{m}^*, \Sigma^*)$, define $\mathsf{FreshID}$ as the event that $\mathcal{A}$ never queries $\mathcal{O}_{\mathsf{Sign}}$ on inputs of the form $(\sigma^*, \cdot)$. Conversely, let $\neg\mathsf{FreshID}$ denote the event that $\mathcal{A}$ ever queried $\mathcal{O}_{\mathsf{Sign}}$ on $(\sigma^*, \cdot)$. If $\neg\mathsf{FreshID}$ occurs, define i_f as follows:

$$i_f := \text{ the first query index w.r.t. } \sigma^* \text{(among the } Q_s \text{ queries to } \mathcal{O}_{\mathsf{Sign}}).$$

$4''$. On receiving a forgery $(\sigma^*, \mathbf{m}^*, \Sigma^*)$ from $\mathcal{A}$, C proceeds as follows.
- If $\sigma^* \in Q_{\mathsf{KeyGen}}$ or $(\sigma^*, \mathbf{m}^*) \in Q_{\mathsf{Sign}}$, G_4 outputs 0.
- Parse $\Sigma^* = (\mathbf{s}^*, \hat{\mathbf{r}}^*, \tilde{\mathbf{r}}^*, \mathbf{z}^*)$ and set $\mathbf{m}^{*\prime} := \mathbf{m}^* \| \mathbf{z}^*$.
- **Forgery-rejection rule 1**: Compute $\hat{\mathbf{r}} := \widehat{\mathsf{PRF}}(\tilde{\mathbf{k}}, \sigma^*)$ and $\tilde{\mathbf{r}} := \widetilde{\mathsf{PRF}}(\tilde{\mathbf{k}}, \sigma^* \| \mathbf{m}^{*\prime})$. If $\hat{\mathbf{r}}^* = \hat{\mathbf{r}}$ or $\tilde{\mathbf{r}}^* = \tilde{\mathbf{r}}$, G_4 outputs 0.
- **Forgery-rejection rule 2**: If $\mathsf{FreshID} \wedge i^* \neq Q_s + 1$ or $\neg\mathsf{FreshID} \wedge i^* \neq i_f$, G_4 outputs 0.
- Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma^*), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i \in [\hat{l}]})$.
- Construct $f_{\hat{\mathbf{r}}^*}$ and compute $\mathbf{C}_{f_{\hat{\mathbf{r}}^*}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}^*}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i \in [\hat{y}]})$.
- Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widetilde{\mathsf{PRF}}(\cdot, \sigma^* \| \mathbf{m}^{*\prime}), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i \in [\tilde{l}]})$.
- Construct $f_{\tilde{\mathbf{r}}^*}$ and compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}^*}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i \in [\tilde{y}]})$.
- If $[\mathbf{A} | \mathbf{C}_{f_{\hat{\mathbf{r}}^*}} | \mathbf{C}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^* = \mathbf{0}^n$ and $0 < \|\mathbf{s}^*\| \leq \delta\sqrt{m + 2w}$, G_4 outputs 1; otherwise, G_4 outputs 0.

Lemma 14. $\Pr[\mathsf{G}_4(\kappa) \Rightarrow 1] = \frac{1}{Q_s+1} \Pr[\mathsf{G}_3(\kappa) \Rightarrow 1]$.

Proof of Lemma 14. For G_3, we know that $i^* \xleftarrow{\$} [Q_s + 1]$.

$$\Pr[\mathsf{G}_3(\kappa) \Rightarrow 1] = \Pr[\mathsf{G}_3(\kappa) \Rightarrow 1 \wedge \mathsf{FreshID}] + \Pr[\mathsf{G}_3(\kappa) \Rightarrow 1 \wedge \neg\mathsf{FreshID}].$$

For G_4, we know that

$$\begin{aligned}
\Pr[\mathsf{G}_4(\kappa) \Rightarrow 1] &= \Pr[\mathsf{G}_3(\kappa) \Rightarrow 1 \wedge \mathsf{FreshID} \wedge i^* = Q_s + 1] \\
&\quad + \Pr[\mathsf{G}_3(\kappa) \Rightarrow 1 \wedge \neg\mathsf{FreshID} \wedge i^* = i_f] \\
&= \frac{1}{Q_s + 1} \Pr[\mathsf{G}_3(\kappa) \Rightarrow 1 \wedge \mathsf{FreshID}] + \frac{1}{Q_s + 1} \Pr[\mathsf{G}_3(\kappa) \Rightarrow 1 \wedge \neg\mathsf{FreshID}] \\
&= \frac{1}{Q_s + 1} \Pr[\mathsf{G}_3(\kappa) \Rightarrow 1]
\end{aligned}$$

The first equation is due to the rejection rule, and the second equation is due to the independence between i^* and $\mathcal{A}$'s winning condition and the fact that $i^* \xleftarrow{\$} [Q_s + 1]$. □

Game G_5**.** Recall that σ_{i^*} denotes the identity extracted from adversary $\mathcal{A}$'s i^*th query to the signing oracle if $i^* \in [Q_s]$. Game G_5 is identical to G_4 except for the reply to the i^*th signing query $(\sigma_{i^*}, \mathbf{m})$ and the later signing queries $(\sigma = \sigma_{i^*}, \cdot)$.

- For the i^*th signing query $(\sigma_{i^*}, \mathbf{m})$, game G_5 aborts by outputting 0 if this is not the first signing query for the identity σ_{i^*}, i.e., $\exists (sk_{\sigma_{i^*}}, \sigma_{i^*}) \in Q_{sk}$.
- For the later (than i^*) signing queries $(\sigma = \sigma_{i^*}, \cdot)$, game G_5 computes the trapdoor $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}$ by a G-trapdoor of $[\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}]$, rather than using $\mathbf{T}_{\mathbf{A}}$ to compute $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}}$ and then to compute $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}$.

More precisely, step $3'$ is changed to $3''$.

$3''$. On receiving a signing query $(\sigma, \mathbf{m})$ from $\mathcal{A}$, C proceeds as follows.
- Add $(\sigma, \mathbf{m})$ to Q_{Sign} and update $\mathsf{cnt} := \mathsf{cnt} + 1$.
- Sample $\mathbf{z} \xleftarrow{\$} \{0,1\}^\kappa$ and set $\mathbf{m}' := \mathbf{m} \| \mathbf{z}$.
- If $\mathsf{cnt} = i^*$, set $\sigma_{i^*} := \sigma$.
- If $\mathsf{cnt} = i^*$ and $\exists (sk_{\sigma_{i^*}}, \sigma_{i^*}) \in Q_{sk}$, G_5 aborts by outputting 0.
- If $\mathsf{cnt} = i^*$ and $\nexists (sk_{\sigma_{i^*}}, \sigma_{i^*}) \in Q_{sk}$, sample $\hat{\mathbf{r}}_{i^*} \xleftarrow{\$} \{0,1\}^{\hat{y}}$.
- If $\sigma = \sigma_{i^*}$:
  - Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i \in [\hat{l}]})$ and construct $f_{\hat{\mathbf{r}}_{i^*}}$ as defined by (3.2).
  - Compute $\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}_{i^*}}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i \in [\hat{y}]})$.
  - Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widetilde{\mathsf{PRF}}(\cdot, \sigma \| \mathbf{m}'), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i \in [\tilde{l}]})$ and $(\tilde{\mathbf{R}}'_1, \ldots, \tilde{\mathbf{R}}'_{\tilde{y}}) \leftarrow \mathsf{Eval}_{prv}(\widetilde{\mathsf{PRF}}(\cdot, \sigma \| \mathbf{m}'), \mathbf{A}, \tilde{\mathbf{k}}, \{\tilde{\mathbf{R}}_i\}_{i \in [\tilde{l}]})$.
  - Compute $\tilde{\mathbf{r}} := \widetilde{\mathsf{PRF}}(\tilde{\mathbf{k}}, \sigma \| \mathbf{m}')$. Construct $f_{\tilde{\mathbf{r}}}$ as defined by (3.2).
  - Compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i \in [\tilde{y}]})$ and $\mathbf{R}_{f_{\tilde{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{prv}(f_{\tilde{\mathbf{r}}}(\cdot), \mathbf{A}, \tilde{\mathbf{r}}, \{\tilde{\mathbf{R}}'_i\}_{i \in [\tilde{y}]})$.
  - Generate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}} \leftarrow \mathsf{GtoBasis}(\mathbf{R}_{f_{\tilde{\mathbf{r}}}})$.
  - Delegate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}})$.
  - Sample $\mathbf{s} \leftarrow \mathsf{SamplePre}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}, \mathbf{0}^n, \delta)$ with $\mathbf{s} \in \mathbb{Z}_q^{m+2w}$ and $\mathbf{s} \neq \mathbf{0}^{m+2w}$.

- Return $\Sigma = (\mathbf{s}, \hat{\mathbf{r}}_{i^*}, \tilde{\mathbf{r}}, \mathbf{z})$.

– If $\sigma \neq \sigma_{i^*}$:
- If $\nexists (sk_\sigma, \sigma) \in \mathcal{Q}_{sk}$, then generate sk_σ by invoking the key generation query on σ as in $\mathcal{O}_{\mathsf{KeyGen}}$.
- Parse $sk_\sigma = (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}}, \mathbf{C}_{f_{\hat{\mathbf{r}}}}, \sigma)$.
- Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma \| \mathbf{m}'), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{l}]})$.
- Sample $\tilde{\mathbf{r}} \leftarrow \{0,1\}^{\tilde{y}}$. Construct $f_{\tilde{\mathbf{r}}}$ as defined by (3.2).
- Compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i\in[\tilde{y}]})$.
- Delegate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}})$.
- Sample $\mathbf{s} \leftarrow \mathsf{SamplePre}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}, \mathbf{0}^n, \delta)$ with $\mathbf{s} \in \mathbb{Z}_q^{m+2w}$ and $\mathbf{s} \neq \mathbf{0}^{m+2w}$.
- Return $\Sigma = (\mathbf{s}, \hat{\mathbf{r}}, \tilde{\mathbf{r}}, \mathbf{z})$.

Lemma 15. $|\Pr[\mathsf{G}_5(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_4(\kappa) \Rightarrow 1]| \leq 2^{-\kappa}$.

Proof of Lemma 15. If $\mathsf{cnt} = i^*$ and $\exists (sk_{\sigma^*}, \sigma^*) \in \mathcal{Q}_{sk}$ in G_4, then adversary $\mathcal{A}$ has queried $\mathcal{O}_{\mathsf{KeyGen}}$ with σ^* or $(\sigma^*, \mathbf{m})$ is not the first query to $\mathcal{O}_{\mathsf{Sign}}$ w.r.t. σ^*. In this case, either $\sigma^* \in \mathcal{Q}_{\mathsf{KeyGen}}$ or **rejection-rule 2** is triggered, and thus G_4 outputs 0. Therefore, G_4 and G_5 behave the same when $\sigma \neq \sigma_{i^*}$.

Hence the only difference between G_4 and G_5 is the generation of $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}$ when answering the signing queries $(\sigma = \sigma_{i^*}, \cdot)$. In G_4, $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}$ is computed by $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}})$ where $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}}$ is derived from $\mathbf{T}_{\mathbf{A}}$. In comparison, in G_5, $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}$ is derived from $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}$ which is computed from the G-trapdoor of $[\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}]$. By Lemma 7, we have $\mathbf{C}_{f_{\tilde{\mathbf{r}}}} = \mathbf{A}\mathbf{R}_{f_{\tilde{\mathbf{r}}}} + f_{\tilde{\mathbf{r}}}(\tilde{\mathbf{r}}) \cdot \mathbf{G} = \mathbf{A}\mathbf{R}_{f_{\tilde{\mathbf{r}}}} + \mathbf{G}$, so $\mathbf{R}_{f_{\tilde{\mathbf{r}}}}$ is a G-trapdoor of $[\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}]$. Hence $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}} \leftarrow \mathsf{GtoBasis}(\mathbf{R}_{f_{\tilde{\mathbf{r}}}})$ is a trapdoor for $[\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}]$ with norm smaller than $O(\kappa^c)$ for some constant c according to Lemma 5. By Lemma 3 the delegated trapdoor satisfies $\|\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}}\| = \|\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}\|$. Finally, combining Lemma 2 and $\delta \geq O(\kappa^c) \cdot \omega(\sqrt{\log m})$, we conclude that the distributions of $\mathbf{s}$ generated in G_4 and G_5 are statistically indistinguishable. As a consequence the resulting signatures $\Sigma = (\mathbf{s}, \hat{\mathbf{r}}_{i^*}, \tilde{\mathbf{r}}, \mathbf{z})$ in G_4 and G_5 are also statistically indistinguishable. Therefore Lemma 15 follows. □

Game G_6**.** Game G_6 is identical to G_5 except for the generation of $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}$ in key generation oracle $\mathcal{O}_{\mathsf{KeyGen}}$. In G_6, if the query satisfies $\sigma = \sigma_{i^*}$, game G_6 directly outputs 0; otherwise, the trapdoor $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}$ is derived from a G-trapdoor of $[\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}]$ instead of being computed from $\mathbf{T}_{\mathbf{A}}$ as in G_5.

$2''$. On receiving a key generation query σ from $\mathcal{A}$, $\mathcal{C}$ proceeds as follows.
– Add σ to $\mathcal{Q}_{\mathsf{KeyGen}}$. If $\exists (sk_\sigma, \sigma) \in \mathcal{Q}_{sk}$, return sk_σ.
– If $\sigma = \sigma_{i^*}$, game G_6 aborts by outputting 0.
– Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{l}]})$ and $(\hat{\mathbf{R}}'_1, \ldots, \hat{\mathbf{R}}'_{\hat{y}}) \leftarrow \mathsf{Eval}_{prv}(\widehat{\mathsf{PRF}}(\cdot, \sigma), \mathbf{A}, \hat{\mathbf{k}}, \{\hat{\mathbf{R}}_i\}_{i\in[\hat{l}]})$.
– Compute $\hat{\mathbf{r}} := \widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma)$ and construct $f_{\hat{\mathbf{r}}}$ as defined by (3.2).
– Compute $\mathbf{C}_{f_{\hat{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i\in[\hat{y}]})$ and $\mathbf{R}_{f_{\hat{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{prv}(f_{\hat{\mathbf{r}}}(\cdot), \mathbf{A}, \hat{\mathbf{r}}, \{\hat{\mathbf{R}}'_i\}_{i\in[\hat{y}]})$.
– Generate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{GtoBasis}(\mathbf{R}_{f_{\hat{\mathbf{r}}}})$ and randomize it by $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{RandBasis}(\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \gamma)$.
– Return $sk_\sigma = (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}}, \mathbf{C}_{f_{\hat{\mathbf{r}}}}, \sigma)$ to $\mathcal{A}$ and add (sk_σ, σ) to $\mathcal{Q}_{sk}$.

Lemma 16. $|\Pr[\mathsf{G}_6(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_5(\kappa) \Rightarrow 1]| \leq 2^{-\kappa}$.

Proof of Lemma 16. Due to rejection rule 2, G_5 outputting 1 implies two facts: (i) if $i^* \in [Q_s]$, then $\sigma^* = \sigma_{i^*}$ and i^* is the index of the first signing query w.r.t σ^*; (ii) σ^* was never submitted to the key generation oracle. In G_5, if a key generation query results in $\sigma = \sigma_{i^*}$, then either $\sigma^* \in \mathcal{Q}_{\mathsf{KeyGen}}$ or **rejection rule 2** is triggered and thus G_5 outputs 0. Consequently, G_5 and G_6 behave the same in this case.

So the only difference between G_5 and G_6 is the generation of the trapdoor $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}$ for $[\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}]$ when $\sigma \neq \sigma_{i^*}$. In G_5, the trapdoor $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}$ is generated by $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}})$ and then randomized by $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{RandBasis}(\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \gamma)$. In G_6, it is generated by $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{GtoBasis}(\mathbf{R}_{f_{\hat{\mathbf{r}}}})$ and then randomized by $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{RandBasis}(\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \gamma)$. According to Lemma 7, we know $\mathbf{C}_{f_{\hat{\mathbf{r}}}} = \mathbf{A}\mathbf{R}_{f_{\hat{\mathbf{r}}}} + f_{\hat{\mathbf{r}}}(\hat{\mathbf{r}}) \cdot \mathbf{G} = \mathbf{A}\mathbf{R}_{f_{\hat{\mathbf{r}}}} + \mathbf{G}$. Therefore, $\mathbf{R}_{f_{\hat{\mathbf{r}}}}$ is a G-trapdoor of $[\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}]$, and hence $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{GtoBasis}(\mathbf{R}_{f_{\hat{\mathbf{r}}}})$ is also a trapdoor for $[\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}]$ with norm smaller than $O(\kappa^c)$ for some constant c according to Lemma 5. Given $\gamma \geq O(\kappa^c) \cdot \omega(\sqrt{n})$, Lemma 4 suggests that the distributions of $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}$ generated in G_5 and G_6 are statistically indistinguishable. □

Game G_7**.** Game G_7 is identical to G_6 except for the generation of $\mathbf{A}$. In G_7, $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$ is sampled uniformly at random instead of being generated by algorithm $\mathsf{TrapGen}$.

$1''$. During the setup phase, challenger $\mathcal{C}$ proceeds as follows.
– Generate $\hat{pp} \leftarrow \widehat{\mathsf{PRF}}.\mathsf{Setup}(1^\kappa)$ and $\tilde{pp} \leftarrow \widehat{\mathsf{PRF}}.\mathsf{Setup}(1^\kappa)$.

– Generate $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$.
– Sample $\hat{\mathbf{R}}_i \xleftarrow{\$} \{1,-1\}^{m\times w}$ and set $\hat{\mathbf{A}}_i := \mathbf{A}\hat{\mathbf{R}}_i + \hat{k}_i\mathbf{G}$ for $i \in [\hat{t}]$, sample $\tilde{\mathbf{R}}_i \xleftarrow{\$} \{1,-1\}^{m\times w}$ and set $\tilde{\mathbf{A}}_i := \mathbf{A}\tilde{\mathbf{R}}_i + \tilde{k}_i\mathbf{G}$ for $i \in [\tilde{t}]$.
– Set $mvk = (\mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]}, \hat{pp}, \tilde{pp})$ and $msk = (mvk, \perp, \hat{\mathbf{k}}, \tilde{\mathbf{k}})$ and return mvk to $\mathcal{A}$.

Lemma 17. $|\Pr[\mathsf{G}_7(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_6(\kappa) \Rightarrow 1]| \leq 2^{-\kappa}$.

Proof of Lemma 17. By Lemma 1, we know that $\mathsf{TrapGen}(1^n, 1^m, q)$ outputs a uniform $\mathbf{A}$ except with probability $2^{-n} \leq 2^{-\kappa}$. Therefore, Lemma 17 holds. □

Lemma 18. $\Pr[\mathsf{G}_7(\kappa) \Rightarrow 1] \leq \mathsf{Adv}^{\mathsf{SIS}}_{[n,q,\beta,m]}(\kappa)$.

Proof of Lemma 18. To prove this, we show that if there exists a PPT adversary $\mathcal{A}$ that wins in G_7, then there exists a PPT algorithm $\mathcal{B}_3$ that solves the $\mathsf{SIS}_{n,q,\beta,m}$ problem. The algorithm $\mathcal{B}_3$ is described below.
Algorithm $\mathcal{B}_3$. Algorithm $\mathcal{B}_3$ is given an $\mathsf{SIS}_{n,q,\beta,m}$ instance $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ and aims to output a vector $\mathbf{v}$ s.t. $\mathbf{A}\mathbf{v} = \mathbf{0}^n$, $\mathbf{v} \neq \mathbf{0}^m$ and $\|\mathbf{v}\| \leq \beta$. To this end, $\mathcal{B}_3$ simulates G_7 for $\mathcal{A}$ as follows.

0. The algorithm $\mathcal{B}_3$ initializes sets $\mathcal{Q}_{\mathsf{KeyGen}} := \emptyset$, $\mathcal{Q}_{\mathsf{Sign}} := \emptyset$, $\mathcal{Q}_{sk} := \emptyset$, samples $i^* \xleftarrow{\$} [Q_s + 1]$, samples $\hat{\mathbf{k}} \xleftarrow{\$} \{0,1\}^{\hat{l}}$, $\tilde{\mathbf{k}} \xleftarrow{\$} \{0,1\}^{\tilde{l}}$ as secret keys for $\widehat{\mathsf{PRF}}$ and $\widetilde{\mathsf{PRF}}$ and initializes the counter $\mathsf{cnt} := 0$ and identity $\sigma_{i^*} := \perp$.
1. During the setup phase, algorithm $\mathcal{B}_3$ proceeds as follows.
   – Generate $\hat{pp} \leftarrow \widehat{\mathsf{PRF}}.\mathsf{Setup}(1^\kappa)$ and $\tilde{pp} \leftarrow \widetilde{\mathsf{PRF}}.\mathsf{Setup}(1^\kappa)$.
   – Sample $\hat{\mathbf{R}}_i \xleftarrow{\$} \{1,-1\}^{m\times w}$ and set $\hat{\mathbf{A}}_i := \mathbf{A}\hat{\mathbf{R}}_i + \hat{k}_i\mathbf{G}$ for $i \in [\hat{t}]$, sample $\tilde{\mathbf{R}}_i \xleftarrow{\$} \{1,-1\}^{m\times w}$ and set $\tilde{\mathbf{A}}_i := \mathbf{A}\tilde{\mathbf{R}}_i + \tilde{k}_i\mathbf{G}$ for $i \in [\tilde{t}]$ (here $\mathbf{A}$ is the SIS instance).
   – Set $mvk = (\mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]}, \hat{pp}, \tilde{pp})$ and $msk = (mvk, \perp, \hat{\mathbf{k}}, \tilde{\mathbf{k}})$ and return mvk to $\mathcal{A}$.
2. On receiving a key generation query σ from $\mathcal{A}$, $\mathcal{B}_3$ proceeds as follows.
   – Add σ to $\mathcal{Q}_{\mathsf{KeyGen}}$. If $\exists(sk_\sigma, \sigma) \in \mathcal{Q}_{sk}$, return sk_σ.
   – If $\sigma = \sigma_{i^*}$, $\mathcal{B}_3$ aborts by outputting 0.
   – Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]})$ and $(\hat{\mathbf{R}}'_1, \ldots, \hat{\mathbf{R}}'_{\hat{y}}) \leftarrow \mathsf{Eval}_{prv}(\widehat{\mathsf{PRF}}(\cdot, \sigma), \mathbf{A}, \hat{\mathbf{k}}, \{\hat{\mathbf{R}}_i\}_{i\in[\hat{t}]})$.
   – Compute $\hat{\mathbf{r}} := \widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma)$ and construct $f_{\hat{\mathbf{r}}}$ as defined by (3.2).
   – Compute $\mathbf{C}_{f_{\hat{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i\in[\hat{y}]})$ and $\mathbf{R}_{f_{\hat{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{prv}(f_{\hat{\mathbf{r}}}(\cdot), \mathbf{A}, \hat{\mathbf{r}}, \{\hat{\mathbf{R}}'_i\}_{i\in[\hat{y}]})$.
   – Generate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{GtoBasis}(\mathbf{R}_{f_{\hat{\mathbf{r}}}})$ and randomize it by $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}} \leftarrow \mathsf{RandBasis}(\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \gamma)$.
   – Return $sk_\sigma = (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}}, \mathbf{C}_{f_{\hat{\mathbf{r}}}}, \sigma)$ to $\mathcal{A}$ and add (sk_σ, σ) to $\mathcal{Q}_{sk}$.
3. On receiving a signing query $(\sigma, \mathbf{m})$ from $\mathcal{A}$, $\mathcal{B}_3$ proceeds as follows.
   – Add $(\sigma, \mathbf{m})$ to $\mathcal{Q}_{\mathsf{Sign}}$ and update $\mathsf{cnt} := \mathsf{cnt} + 1$.
   – Sample $\mathbf{z} \xleftarrow{\$} \{0,1\}^\kappa$ and set $\mathbf{m}' := \mathbf{m}\|\mathbf{z}$.
   – If $\mathsf{cnt} = i^*$, set $\sigma_{i^*} := \sigma$.
   – If $\mathsf{cnt} = i^*$ and $\exists(sk_{\sigma_{i^*}}, \sigma_{i^*}) \in \mathcal{Q}_{sk}$, $\mathcal{B}_3$ aborts by outputting 0.
   – If $\mathsf{cnt} = i^*$ and $\nexists(sk_{\sigma_{i^*}}, \sigma_{i^*}) \in \mathcal{Q}_{sk}$, sample $\hat{\mathbf{r}}_{i^*} \xleftarrow{\$} \{0,1\}^{\hat{y}}$.
   – If $\sigma = \sigma_{i^*}$:
     • Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{t}]})$ and construct $f_{\hat{\mathbf{r}}_{i^*}}$ as defined by (3.2).
     • Compute $\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}_{i^*}}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i\in[\hat{y}]})$.
     • Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widetilde{\mathsf{PRF}}(\cdot, \sigma\|\mathbf{m}'), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]})$ and $(\tilde{\mathbf{R}}'_1, \ldots, \tilde{\mathbf{R}}'_{\tilde{y}}) \leftarrow \mathsf{Eval}_{prv}(\widetilde{\mathsf{PRF}}(\cdot, \sigma\|\mathbf{m}'), \mathbf{A}, \tilde{\mathbf{k}}, \{\tilde{\mathbf{R}}_i\}_{i\in[\tilde{t}]})$.
     • Compute $\tilde{\mathbf{r}} := \widetilde{\mathsf{PRF}}(\tilde{\mathbf{k}}, \sigma\|\mathbf{m}')$. Construct $f_{\tilde{\mathbf{r}}}$ as defined by (3.2).
     • Compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i\in[\tilde{y}]})$ and $\mathbf{R}_{f_{\tilde{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{prv}(f_{\tilde{\mathbf{r}}}(\cdot), \mathbf{A}, \tilde{\mathbf{r}}, \{\tilde{\mathbf{R}}'_i\}_{i\in[\tilde{y}]})$.
     • Generate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}} \leftarrow \mathsf{GtoBasis}(\mathbf{R}_{f_{\tilde{\mathbf{r}}}})$.
     • Delegate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}})$ and compute $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}$.
     • Sample $\mathbf{s} \leftarrow \mathsf{SamplePre}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}_{i^*}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}, \mathbf{0}^n, \delta)$ with $\mathbf{s} \in \mathbb{Z}_q^{m+2w}$ and $\mathbf{s} \neq \mathbf{0}^{m+2w}$.
     • Return $\Sigma = (\mathbf{s}, \hat{\mathbf{r}}_{i^*}, \tilde{\mathbf{r}}, \mathbf{z})$.
   – If $\sigma \neq \sigma_{i^*}$:
     • If $\nexists(sk_\sigma, \sigma) \in \mathcal{Q}_{sk}$, then generate sk_σ by invoking the key generation query on σ as in $\mathcal{O}_{\mathsf{KeyGen}}$.
     • Parse $sk_\sigma = (\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}}, \hat{\mathbf{r}}, \mathbf{C}_{f_{\hat{\mathbf{r}}}}, \sigma)$.
     • Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widetilde{\mathsf{PRF}}(\cdot, \sigma\|\mathbf{m}'), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{t}]})$.
     • Sample $\tilde{\mathbf{r}} \leftarrow \{0,1\}^{\tilde{y}}$. Construct $f_{\tilde{\mathbf{r}}}$ as defined by (3.2).
     • Compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i\in[\tilde{y}]})$.
     • Delegate $\mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}} \leftarrow \mathsf{DelTrap}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}})$.
     • Sample $\mathbf{s} \leftarrow \mathsf{SamplePre}([\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}], \mathbf{T}_{\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}}}}, \mathbf{0}^n, \delta)$ with $\mathbf{s} \in \mathbb{Z}_q^{m+2w}$ and $\mathbf{s} \neq \mathbf{0}^{m+2w}$.
     • Return $\Sigma = (\mathbf{s}, \hat{\mathbf{r}}, \tilde{\mathbf{r}}, \mathbf{z})$.
4. On receiving a forgery $(\sigma^*, \mathbf{m}^*, \Sigma^*)$ from $\mathcal{A}$, $\mathcal{B}_3$ proceeds as follows.

- If $\sigma^* \in Q_{\mathsf{KeyGen}}$ or $(\sigma^*, \mathbf{m}^*) \in Q_{\mathsf{Sign}}$, return 0.
- Parse $\Sigma^* = (\mathbf{s}^*, \hat{\mathbf{r}}^*, \tilde{\mathbf{r}}^*, \mathbf{z}^*)$ and set $\mathbf{m}^{*\prime} := \mathbf{m}^* \| \mathbf{z}^*$.
- **Forgery-rejection rule 1**: Compute $\hat{\mathbf{r}} := \widehat{\mathsf{PRF}}(\hat{\mathbf{k}}, \sigma^*)$ and $\tilde{\mathbf{r}} := \widetilde{\mathsf{PRF}}(\tilde{\mathbf{k}}, \sigma^* \| \mathbf{m}^{*\prime})$. If $\hat{\mathbf{r}}^* = \hat{\mathbf{r}}$ or $\tilde{\mathbf{r}}^* = \tilde{\mathbf{r}}$, return 0.
- **Forgery-rejection rule 2**: If $\mathsf{FreshID} \wedge i^* \neq Q_s + 1$ or $\neg\mathsf{FreshID} \wedge i^* \neq i_f$, return 0.
- Compute $(\hat{\mathbf{C}}_1, \ldots, \hat{\mathbf{C}}_{\hat{y}}) \leftarrow \mathsf{Eval}_{pub}(\widehat{\mathsf{PRF}}(\cdot, \sigma^*), \mathbf{A}, \{\hat{\mathbf{A}}_i\}_{i\in[\hat{l}]})$ and $(\hat{\mathbf{R}}'_1, \ldots, \hat{\mathbf{R}}'_{\hat{y}}) \leftarrow \mathsf{Eval}_{prv}(\widehat{\mathsf{PRF}}(\cdot, \sigma^*), \mathbf{A}, \hat{\mathbf{k}}, \{\hat{\mathbf{R}}_i\}_{i\in[\hat{l}]})$.
- Construct $f_{\hat{\mathbf{r}}^*}$ and compute $\mathbf{C}_{f_{\hat{\mathbf{r}}^*}} \leftarrow \mathsf{Eval}_{pub}(f_{\hat{\mathbf{r}}^*}(\cdot), \mathbf{A}, \{\hat{\mathbf{C}}_i\}_{i\in[\hat{y}]})$ and $\mathbf{R}_{f_{\hat{\mathbf{r}}^*}} \leftarrow \mathsf{Eval}_{prv}(f_{\hat{\mathbf{r}}^*}(\cdot), \mathbf{A}, \hat{\mathbf{r}}, \{\hat{\mathbf{R}}'_i\}_{i\in[\hat{y}]})$.
- Compute $(\tilde{\mathbf{C}}_1, \ldots, \tilde{\mathbf{C}}_{\tilde{y}}) \leftarrow \mathsf{Eval}_{pub}(\widetilde{\mathsf{PRF}}(\cdot, \sigma^* \| \mathbf{m}^{*\prime}), \mathbf{A}, \{\tilde{\mathbf{A}}_i\}_{i\in[\tilde{l}]})$ and $(\tilde{\mathbf{R}}'_1, \ldots, \tilde{\mathbf{R}}'_{\tilde{y}}) \leftarrow \mathsf{Eval}_{prv}(\widetilde{\mathsf{PRF}}(\cdot, \sigma^* \| \mathbf{m}^{*\prime}), \mathbf{A}, \tilde{\mathbf{k}}, \{\tilde{\mathbf{R}}_i\}_{i\in[\tilde{l}]})$.
- Construct $f_{\tilde{\mathbf{r}}^*}$, compute $\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}} \leftarrow \mathsf{Eval}_{pub}(f_{\tilde{\mathbf{r}}^*}(\cdot), \mathbf{A}, \{\tilde{\mathbf{C}}_i\}_{i\in[\tilde{y}]})$ and $\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}} \leftarrow \mathsf{Eval}_{prv}(f_{\tilde{\mathbf{r}}^*}(\cdot), \mathbf{A}, \tilde{\mathbf{r}}, \{\tilde{\mathbf{R}}'_i\}_{i\in[\tilde{y}]})$.
- If $[\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}^*}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^* \neq \mathbf{0}^n$ or $\|\mathbf{s}^*\| > \delta\sqrt{m+2w}$ or $\mathbf{s}^* = \mathbf{0}^{m+2w}$, return 0; otherwise, compute $\mathbf{v} = [\mathbf{I}|\mathbf{R}_{f_{\hat{\mathbf{r}}^*}}|\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^*$ and return $\mathbf{v}$ to its challenger as a solution to the $\mathsf{SIS}_{n,q,\beta,m}$ problem.

Now we show that $\mathcal{B}_3$ wins if $\mathcal{A}$ wins. If $\mathcal{A}$ wins, then it must hold that $\hat{\mathbf{r}}^* \neq \hat{\mathbf{r}}$, $\tilde{\mathbf{r}}^* \neq \tilde{\mathbf{r}}$, $\mathbf{s}^* \neq \mathbf{0}^{m+2w}$ with $\|\mathbf{s}^*\| \leq \delta\sqrt{m+2w}$ and the following equation holds:

$$\mathbf{0}^n = [\mathbf{A}|\mathbf{C}_{f_{\hat{\mathbf{r}}^*}}|\mathbf{C}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^* = [\mathbf{A}|\mathbf{A}\mathbf{R}_{f_{\hat{\mathbf{r}}^*}} + f_{\hat{\mathbf{r}}^*}(\hat{\mathbf{r}}) \cdot \mathbf{G}|\mathbf{A}\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}} + f_{\tilde{\mathbf{r}}^*}(\tilde{\mathbf{r}}) \cdot \mathbf{G}] \cdot \mathbf{s}^*.$$

Here $\mathbf{C}_{f_{\hat{\mathbf{r}}^*}}, \mathbf{C}_{f_{\tilde{\mathbf{r}}^*}}, \mathbf{R}_{f_{\hat{\mathbf{r}}^*}}, \mathbf{R}_{f_{\tilde{\mathbf{r}}^*}}, \hat{\mathbf{r}}$ and $\tilde{\mathbf{r}}$ are computed as described in algorithm $\mathcal{B}_3$. Since $\hat{\mathbf{r}}^* \neq \hat{\mathbf{r}}$ and $\tilde{\mathbf{r}}^* \neq \tilde{\mathbf{r}}$, it follows that $f_{\hat{\mathbf{r}}^*}(\hat{\mathbf{r}}) = 0$ and $f_{\tilde{\mathbf{r}}^*}(\tilde{\mathbf{r}}) = 0$. Hence we obtain

$$\mathbf{0}^n = [\mathbf{A}|\mathbf{A}\mathbf{R}_{f_{\hat{\mathbf{r}}^*}}|\mathbf{A}\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^* = \mathbf{A} \cdot \underbrace{[\mathbf{I}|\mathbf{R}_{f_{\hat{\mathbf{r}}^*}}|\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^*}_{\mathbf{v}\in\mathbb{Z}_q^m}.$$

Moreover, by our choice of parameters, we have

$$\|[\mathbf{I}|\mathbf{R}_{f_{\hat{\mathbf{r}}^*}}|\mathbf{R}_{f_{\tilde{\mathbf{r}}^*}}] \cdot \mathbf{s}^*\| \leq O(\kappa^c) \cdot \delta\sqrt{m+2w} \leq \beta.$$

Thus, we have found a solution $\mathbf{v}$ s.t., $\mathbf{A}\mathbf{v} = \mathbf{0}^n$, $\mathbf{v} \neq \mathbf{0}^m$ and $\|\mathbf{v}\| \leq \beta$, which solves the $\mathsf{SIS}_{n,q,\beta,m}$ problem. This leads to a contradiction. □

Combining Lemmas 11 to 18, we obtain that (see Fig. 6):

$$\begin{aligned}
\Pr[\mathsf{G}_0(\kappa) \Rightarrow 1] &\leq |\Pr[\mathsf{G}_0(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_1(\kappa) \Rightarrow 1]| + |\Pr[\mathsf{G}_1(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_2(\kappa) \Rightarrow 1]| \\
&\quad + |\Pr[\mathsf{G}_2(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_3(\kappa) \Rightarrow 1]| + |\Pr[\mathsf{G}_3(\kappa) \Rightarrow 1]| \\
&\leq (Q_s+1)\Pr[\mathsf{G}_4(\kappa) \Rightarrow 1] + 2\mathsf{Adv}^{pse}_{\widehat{\mathsf{PRF}}}(\kappa) + 2\mathsf{Adv}^{pse}_{\widetilde{\mathsf{PRF}}}(\kappa) + \mathsf{negl}(\kappa) \\
&\leq (Q_s+1)\big(|\Pr[\mathsf{G}_4(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_5(\kappa) \Rightarrow 1]| + |\Pr[\mathsf{G}_5(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_6(\kappa) \Rightarrow 1]| \\
&\quad + |\Pr[\mathsf{G}_6(\kappa) \Rightarrow 1] - \Pr[\mathsf{G}_7(\kappa) \Rightarrow 1]| + \Pr[\mathsf{G}_7(\kappa) \Rightarrow 1]\big) + 2\mathsf{Adv}^{pse}_{\widehat{\mathsf{PRF}}}(\kappa) + 2\mathsf{Adv}^{pse}_{\widetilde{\mathsf{PRF}}}(\kappa) + \mathsf{negl}(\kappa) \\
&\leq (Q_s+1)(\mathsf{Adv}^{\mathsf{SIS}}_{[n,q,\beta,m]}(\kappa) + 2^{O(-\kappa)}) + 2\mathsf{Adv}^{pse}_{\widehat{\mathsf{PRF}}}(\kappa) + 2\mathsf{Adv}^{pse}_{\widetilde{\mathsf{PRF}}}(\kappa) + \mathsf{negl}(\kappa) \\
&= Q_s\mathsf{Adv}^{\mathsf{SIS}}_{[n,q,\beta,m]}(\kappa) + 2\mathsf{Adv}^{pse}_{\widehat{\mathsf{PRF}}}(\kappa) + 2\mathsf{Adv}^{pse}_{\widetilde{\mathsf{PRF}}}(\kappa) + \mathsf{negl}(\kappa).
\end{aligned}$$

□

4. Identity-based matchmaking encryption from lattices

4.1. IB-ME: syntax, ciphertext pseudo-randomness and (new) stronger authenticity

The security notions of privacy and authenticity were formalized in [2,14]. Later, Wang et al. [16] strengthened the authenticity to stronger authenticity. In their security model of stronger authenticity, the challenger uses a fresh encryption key for every encryption query. However, in practice, once a user asks for an encryption key from the key generation center, he/she will store this encryption key and reuse it for encrypting different messages, rather than impractically asking for a fresh encryption key from the key generation center for each encryption operation. Therefore, the stronger authenticity does not capture the real behavior of users (similar to the $\overline{\text{EUF-ID-CMA}}$ security of IBS).

In this work we formalize a new and more reasonable stronger authenticity notion for IB-ME by allowing a user to use the same encryption key for encrypting different messages. Below we first recall the stronger authenticity definition from [16,17] and then introduce our new stronger authenticity definition. As Theorem 4 shows, the stronger authenticity notion does not imply our definition.

Definition 7 (Identity-based matchmaking encryption [2,14]). Let $\mathcal{ID}$ be the identity space, $\mathcal{M}$ be the message space and $\mathcal{CT}$ be the ciphertext space. An identity-based matchmaking encryption (IB-ME) scheme IB-ME consists of five polynomial-time algorithms $(\mathsf{Setup}, \mathsf{SKeyGen}, \mathsf{RKeyGen}, \mathsf{Enc}, \mathsf{Dec})$ defined below.

- $\mathsf{Setup}(1^\kappa)$ takes as input the security parameter κ, and returns the master public key mpk and master secret key msk.

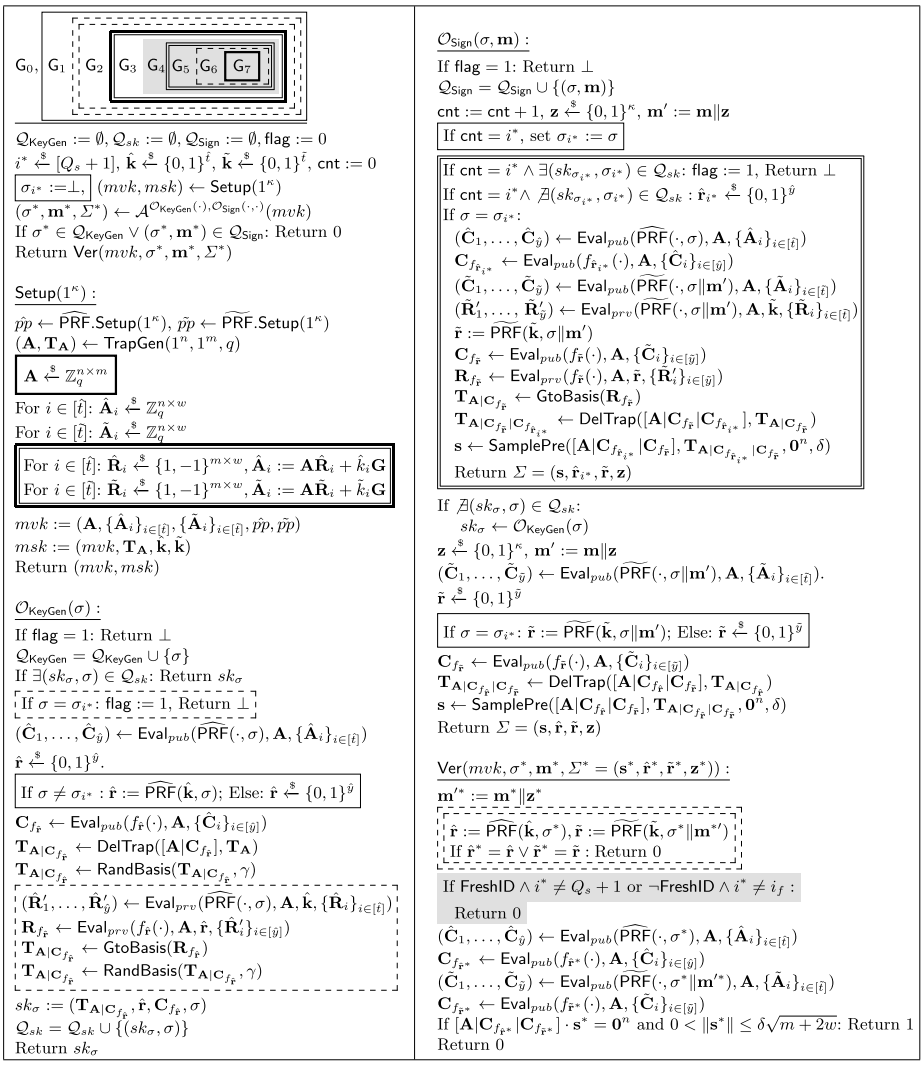


Fig. 6. Game $\mathsf{G}_0 \sim \mathsf{G}_7$ for proving Theorem 3.

- $\mathsf{SKeyGen}(mpk, msk, \sigma)$ takes as input the master public key mpk, master secret key msk and a sender's identity $\sigma \in \mathcal{ID}$, and returns a secret encryption key ek_σ.
- $\mathsf{RKeyGen}(mpk, msk, \rho)$ takes as input the master public key mpk, master secret key msk and a receiver's identity $\rho \in \mathcal{ID}$, and returns a secret decryption key dk_ρ.
- $\mathsf{Enc}(mpk, ek_\sigma, \rho, m)$ takes as input the master public key mpk, an encryption key ek_σ, a target receiver's identity $\rho \in \mathcal{ID}$ and a message $m \in \mathcal{M}$, and returns a ciphertext $c \in \mathcal{CT}$.
- $\mathsf{Dec}(mpk, dk_\rho, \sigma, c)$ takes as input a master public key mpk, a decryption key dk_ρ, a target sender's identity $\sigma \in \mathcal{ID}$ and a ciphertext $c \in \mathcal{CT}$, and returns a message m or $\perp$.

Correctness. For all $\kappa \in \mathbb{N}$ and $(mpk, msk) \leftarrow \mathsf{Setup}(1^\kappa)$, identities $\sigma, \rho \in \mathcal{ID}$, $ek_\sigma \leftarrow \mathsf{SKeyGen}(mpk, msk, \sigma)$ and $dk_\rho \leftarrow \mathsf{RKeyGen}(mpk, msk, \rho)$, for all $m \in \mathcal{M}$, we have:

$$\Pr\left[\mathsf{Dec}(mpk, dk_\rho, \sigma, \mathsf{Enc}(mpk, ek_\sigma, \rho, m)) = m\right] \geq 1 - \mathsf{negl}(\kappa).$$

The following definition captures the ciphertext pseudo-randomness notion for IB-ME; it is analogous to IBE ciphertext pseudo-randomness and stronger than the privacy notion of [2,15,16].

Ciphertext pseudo-randomness. An IB-ME scheme IB-ME satisfies the ciphertext pseudo-randomness if for any PPT adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, its advantage satisfies:

$$|\Pr[\mathsf{Exp}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}}(\kappa) \Rightarrow 1] - 1/2| \leq \mathsf{negl}(\kappa),$$

$\underline{\mathsf{Exp}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}}(\kappa)}$:
$(mpk, msk) \leftarrow \mathsf{Setup}(1^\kappa)$, $\mathcal{Q}_{\mathsf{RKeyGen}} := \emptyset$, $b \xleftarrow{\$} \{0,1\}$
$(\sigma^*, \rho^*, m^*, St) \leftarrow \mathcal{A}_1^{\mathcal{O}_{\mathsf{SKeyGen}}(\cdot),\mathcal{O}_{\mathsf{RKeyGen}}(\cdot)}(mpk)$
$ek_{\sigma^*} \leftarrow \mathsf{SKeyGen}(mpk, msk, \sigma^*)$
$c_0^* \xleftarrow{\$} \mathcal{CT}$, $c_1^* \leftarrow \mathsf{Enc}(mpk, ek_{\sigma^*}, \rho^*, m^*)$
$b' \leftarrow \mathcal{A}_2^{\mathcal{O}_{\mathsf{SKeyGen}}(\cdot),\mathcal{O}_{\mathsf{RKeyGen}}(\cdot)}(c_b^*, ek_{\sigma^*}, St)$
If $\rho^* \in \mathcal{Q}_{\mathsf{RKeyGen}}$: Return b
If $b = b'$: Return 1; Else: Return 0

$\underline{\mathcal{O}_{\mathsf{SKeyGen}}(\sigma)}$:
Return $ek_\sigma \leftarrow \mathsf{SKeyGen}(mpk, msk, \sigma)$

$\underline{\mathcal{O}_{\mathsf{RKeyGen}}(\rho)}$:
$\mathcal{Q}_{\mathsf{RKeyGen}} := \mathcal{Q}_{\mathsf{RKeyGen}} \cup \{\rho\}$
Return $dk_\rho \leftarrow \mathsf{RKeyGen}(mpk, msk, \rho)$

Fig. 7. Experiment $\mathsf{Exp}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}}$ for defining the ciphertext pseudo-randomness of IB-ME.

where the security experiment $\mathsf{Exp}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}}$ is defined in Fig. 7.[1]

Remark 2. Here we show a proof sketch that ciphertext pseudo-randomness is stronger than the privacy notion, where the definition of privacy is given in Appendix B. According to our security notion of ciphertext pseudo-randomness for IB-ME, we have

$$\{\ (ek_{\sigma^*}, c) \mid ek_{\sigma^*} \leftarrow \mathsf{SKeyGen}(mpk, msk, \sigma^*), c \leftarrow \mathsf{Enc}(mpk, ek_{\sigma^*}, \rho^*, m^*)\}$$
$$\approx\ \{\ (ek_{\sigma^*}, u) \mid ek_{\sigma^*} \leftarrow \mathsf{SKeyGen}(mpk, msk, \sigma^*), u \xleftarrow{\$} \mathcal{CT}\},$$

where σ^*, ρ^*, m^* are chosen by $\mathcal{A}$. Clearly, $(ek_{\sigma^*}, c) \approx (ek_{\sigma^*}, u) \Rightarrow c \approx u$. By a hybrid argument, we know that $c_0 \approx u \approx c_1$. Therefore, our ciphertext pseudo-randomness is stronger than the privacy of IB-ME.

Stronger authenticity [16]. An IB-ME scheme IB-ME satisfies the stronger authenticity if for any PPT adversary $\mathcal{A}$, its advantage satisfies:

$$\Pr[\mathsf{Exp}^{saut}_{\mathsf{IB\text{-}ME},\mathcal{A}}(\kappa) \Rightarrow 1] \leq \mathsf{negl}(\kappa),$$

where the experiment $\mathsf{Exp}^{saut}_{\mathsf{IB\text{-}ME},\mathcal{A}}(\kappa)$ is defined in Fig. 8.

Note that in their stronger authenticity model (cf. $\mathsf{Exp}^{saut}_{\mathsf{IB\text{-}ME},\mathcal{A}}(\kappa)$), for each encryption query $\mathsf{Enc}(\sigma, \rho, m)$, the challenger generates a *fresh* encryption key ek_σ for user σ, and uses it to generate a ciphertext for message m. However, in practice, it is impractical for a user σ to ask for a fresh encryption key from the key generation center for each encryption procedure. Therefore, the stronger authenticity notion for IB-ME in [16,17] does not capture the essential security of IB-ME.

More reasonably, a user will keep his/her encryption key ek_σ after he/she obtains it from the key generation center. Then the user will use this ek_σ to encrypt every message thereafter. Based on this observation, we define a *new* stronger authenticity under chosen-identity/message attack in Fig. 8.

New stronger authenticity. An IB-ME scheme IB-ME satisfies the new stronger authenticity if for any PPT adversary $\mathcal{A}$, its advantage satisfies:

$$\Pr[\mathsf{Exp}^{new\text{-}saut}_{\mathsf{IB\text{-}ME},\mathcal{A}}(\kappa) \Rightarrow 1] \leq \mathsf{negl}(\kappa),$$

where the experiment $\mathsf{Exp}^{new\text{-}saut}_{\mathsf{IB\text{-}ME},\mathcal{A}}(\kappa)$ is defined in Fig. 8.

4.2. Separation between stronger authenticity and the new notion

If algorithm SKeyGen in IB-ME is deterministic, then the stronger authenticity for IB-ME in [16,17] is equivalent to our new stronger authenticity. Conversely, if algorithm SKeyGen is probabilistic, we show the two security notions are different by providing a separation (analogous to the separation for IBS in Section 3.2).

In [16], Wang et al. proposed a generic construction of IB-ME from anonymous HIBE and IBS, and showed that the stronger authenticity security of IB-ME is reduced to the $\overline{\text{EUF-ID-CMA}}$ of the underlying IBS. We now construct the scheme $\overline{\text{IB-ME}}$ by instantiating the HIBE component with HIBE from [28] and the IBS component with $\overline{\text{IBS}}$ (shown in Section 3.2). Given that $\overline{\text{IBS}}$ has $\overline{\text{EUF-ID-CMA}}$ security, it follows directly from the reduction in [16] that $\overline{\text{IB-ME}}$ achieves stronger authenticity, as shown in Lemma 19.

Lemma 19. *The IB-ME scheme $\overline{\text{IB-ME}}$ in Fig. 9 satisfies stronger authenticity. More precisely, for any PPT adversary $\mathcal{A}$, there exists a PPT algorithm $\mathcal{B}$ s.t.*

$$\mathsf{Adv}^{saut}_{\overline{\text{IB-ME}},\mathcal{A}}(\kappa) \leq \mathsf{Adv}^{\overline{\text{EUF-ID-CMA}}}_{\overline{\text{IBS}},\mathcal{B}}(\kappa) + \mathsf{negl}(\kappa)$$

However, we prove that $\overline{\text{IB-ME}}$ does not have new stronger authenticity, yielding the separation between the stronger authenticity and our new stronger authenticity, as formalized in Theorem 4.

Theorem 4. *The IB-ME scheme $\overline{\text{IB-ME}}$ in Fig. 9 satisfies stronger authenticity but does not satisfy new stronger authenticity.*

[1] Similar to [2], $\mathsf{Exp}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}}$ has to avoid trivial attacks: if an adversary is allowed to obtain dk_{ρ^*} from $\mathcal{O}_{\mathsf{RKeyGen}}$, then it can trivially win the experiment by checking whether $\mathsf{Dec}(mpk, dk_{\rho^*}, \sigma^*, c_b^*)$ is $\perp$ or not.

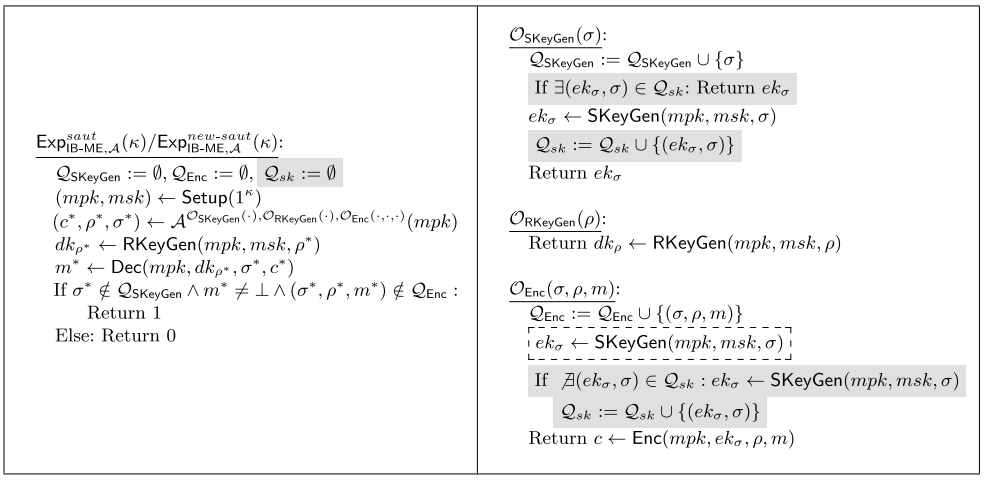


Fig. 8. $\mathsf{Exp}^{saut}_{\overline{\mathsf{IB\text{-}ME}},\mathcal{A}}$ (with the dashed parts) for defining the stronger authenticity of $\overline{\mathsf{IB\text{-}ME}}$ (defined in [16,17]) and Experiment $\mathsf{Exp}^{new\text{-}saut}_{\overline{\mathsf{IB\text{-}ME}},\mathcal{A}}$ (with the gray parts) for defining the new stronger authenticity of $\overline{\mathsf{IB\text{-}ME}}$.

$(mpk, msk) \leftarrow \mathsf{Setup}(1^\kappa)$.
1. $(\mathsf{HIBE}.mpk, \mathsf{HIBE}.msk) \leftarrow \mathsf{HIBE.Setup}(1^\kappa)$.
2. $(\overline{\mathsf{IBS}}.mvk, \overline{\mathsf{IBS}}.msk) \leftarrow \overline{\mathsf{IBS}}.\mathsf{Setup}(1^\kappa)$.
3. Return $mpk := (\mathsf{HIBE}.mpk, \overline{\mathsf{IBS}}.mvk)$ and $msk := (\mathsf{HIBE}.msk, \overline{\mathsf{IBS}}.msk)$.

$ek_\sigma \leftarrow \mathsf{SKeyGen}(mpk, msk, \sigma)$.
1. Parse $msk = (\mathsf{HIBE}.msk, \overline{\mathsf{IBS}}.msk)$.
2. Compute $\overline{\mathsf{IBS}}.sk_\sigma \leftarrow \overline{\mathsf{IBS}}.\mathsf{KeyGen}(\overline{\mathsf{IBS}}.msk, \sigma)$.
3. Return $ek_\sigma := (\overline{\mathsf{IBS}}.sk_\sigma, \sigma)$.

$dk_\rho \leftarrow \mathsf{RKeyGen}(mpk, msk, \rho)$.
1. Parse $msk = (\mathsf{HIBE}.msk, \overline{\mathsf{IBS}}.msk)$.
2. Compute $\mathsf{HIBE}.sk_\rho \leftarrow \mathsf{HIBE.KeyGen}(\mathsf{HIBE}.msk, \rho)$.
3. Return $dk_\rho := (\mathsf{HIBE}.sk_\rho, \rho)$.

$c \leftarrow \mathsf{Enc}(mpk, ek_\sigma, \rho, m)$.
1. Parse $mpk = (\mathsf{HIBE}.mpk, \overline{\mathsf{IBS}}.mvk)$ and $ek_\sigma = (\overline{\mathsf{IBS}}.sk_\sigma, \sigma)$
2. Compute $\Sigma \leftarrow \overline{\mathsf{IBS}}.\mathsf{Sign}(\overline{\mathsf{IBS}}.sk_\sigma, (\rho, m))$.
3. Set $m' := (\Sigma, m)$, return $c \leftarrow \mathsf{HIBE.Enc}(\mathsf{HIBE}.mpk, \rho \| \sigma, m')$.

$m \leftarrow \mathsf{Dec}(mpk, dk_\rho, \sigma, c)$.
1. Parse $mpk = (\mathsf{HIBE}.mpk, \overline{\mathsf{IBS}}.mvk)$ and $dk_\rho = (\mathsf{HIBE}.sk_\rho, \rho)$.
2. Delegate $\mathsf{HIBE}.sk_{\rho\|\sigma} \leftarrow \mathsf{HIBE.Der}(\mathsf{HIBE}.sk_\rho, \rho \| \sigma)$.
3. Compute $m' \leftarrow \mathsf{HIBE.Dec}(\mathsf{HIBE}.mpk, \mathsf{HIBE}.sk_{\rho\|\sigma}, c)$.
4. If $m' = \perp$, return $\perp$. Otherwise, parse $m' = (\Sigma, m)$.
5. If $\overline{\mathsf{IBS}}.\mathsf{Ver}(\overline{\mathsf{IBS}}.mvk, \sigma, (\rho, m), \Sigma) = 1$, return m; otherwise, return $\perp$.

Fig. 9. IB-ME scheme $\overline{\mathsf{IB\text{-}ME}}$ from HIBE and $\overline{\mathsf{IBS}}$ according to [16].

Proof of Theorem 4. By Lemma 19, $\overline{\mathsf{IB\text{-}ME}}$ satisfies stronger authenticity.

We now show that $\overline{\mathsf{IB\text{-}ME}}$ fails to achieve new stronger authenticity by presenting an attack. For the same identities σ and ρ, adversary $\mathcal{A}$ issues two queries (σ, ρ, m) and $(\sigma, \rho, \overline{m})$ to the encryption oracle and obtains the corresponding ciphertexts c and $\tilde{c}$ from the oracle, where $\overline{m}$ denotes the bit-wise complement of m. Subsequently, $\mathcal{A}$ queries the receiver key generation oracle for ρ, obtaining the decryption key dk_ρ. $\mathcal{A}$ then uses dk_ρ to decrypt the ciphertexts c and $\tilde{c}$, yielding the intermediate result $m' = (\Sigma, m)$ and $\tilde{m}' = (\tilde{\Sigma}, \overline{m})$. Due to the correctness of $\overline{\mathsf{IB\text{-}ME}}$, Σ must be a valid $\overline{\mathsf{IBS}}$ signature of (ρ, m) and $\tilde{\Sigma}$ must be a valid $\overline{\mathsf{IBS}}$ signature of $(\rho, \overline{m})$, both generated with respect to σ. By **Fact 2** of $\overline{\mathsf{IBS}}$, the adversary can forge a new signature Σ^* for some $m^* \notin \{m, \overline{m}\}$ under identity σ. Finally, $\mathcal{A}$ encrypts (Σ^*, m^*) to obtain a ciphertext c^* as its forgery via $c^* \leftarrow \mathsf{HIBE.Enc}(\mathsf{HIBE}.mpk, \rho \| \sigma, m' = (\Sigma^*, m^*))$. Clearly, c^* is a valid ciphertext and thus breaks the new stronger authenticity of $\overline{\mathsf{IB\text{-}ME}}$. □

$\underline{(mpk, msk) \leftarrow \mathsf{Setup}(1^\kappa).}$

1. $(\mathsf{IBE}.mpk, \mathsf{IBE}.msk) \leftarrow \mathsf{IBE.Setup}(1^\kappa)$.
2. $(\mathsf{IBS}.mvk, \mathsf{IBS}.msk) \leftarrow \mathsf{IBS.Setup}(1^\kappa)$.
3. Return $mpk := (\mathsf{IBE}.mpk, \mathsf{IBS}.mvk)$ and $msk := (\mathsf{IBE}.msk, \mathsf{IBS}.msk)$.

$\underline{ek_\sigma \leftarrow \mathsf{SKeyGen}(mpk, msk, \sigma).}$

1. Parse $msk = (\mathsf{IBE}.msk, \mathsf{IBS}.msk)$.
2. Compute $\mathsf{IBS}.sk_\sigma \leftarrow \mathsf{IBS.KeyGen}(\mathsf{IBS}.msk, \sigma)$.
3. Return $ek_\sigma := (\mathsf{IBS}.sk_\sigma, \sigma)$.

$\underline{dk_\rho \leftarrow \mathsf{RKeyGen}(mpk, msk, \rho).}$

1. Parse $msk = (\mathsf{IBE}.msk, \mathsf{IBS}.msk)$.
2. Compute $\mathsf{IBE}.sk_\rho \leftarrow \mathsf{IBE.KeyGen}(\mathsf{IBE}.msk, \rho)$.
3. Return $dk_\rho := (\mathsf{IBE}.sk_\rho, \rho)$.

$\underline{c \leftarrow \mathsf{Enc}(mpk, ek_\sigma, \rho, m).}$

1. Parse $mpk = (\mathsf{IBE}.mpk, \mathsf{IBS}.mvk)$ and $ek_\sigma = (\mathsf{IBS}.sk_\sigma, \sigma)$
2. Compute $\Sigma \leftarrow \mathsf{IBS.Sign}(\mathsf{IBS}.sk_\sigma, (\sigma, \rho, m))$.
3. Set $m' := (\Sigma, m)$, return $c \leftarrow \mathsf{IBE.Enc}(\mathsf{IBE}.mpk, \rho, m')$.

$\underline{m \leftarrow \mathsf{Dec}(mpk, dk_\rho, \sigma, c).}$

1. Parse $mpk = (\mathsf{IBE}.mpk, \mathsf{IBS}.mvk)$ and $dk_\rho = (\mathsf{IBE}.sk_\rho, \rho)$.
2. Compute $m' \leftarrow \mathsf{IBE.Dec}(\mathsf{IBE}.mpk, \mathsf{IBE}.sk_\rho, c)$.
3. If $m' = \perp$, return $\perp$. Otherwise, parse $m' = (\Sigma, m)$.
4. If $\mathsf{IBS.Ver}(\mathsf{IBS}.mvk, \sigma, (\sigma, \rho, m), \Sigma) = 1$, return m; otherwise, return $\perp$.

Fig. 10. Generic IB-ME from IBE and IBS.

4.3. Generic IB-ME from IBS and IBE

In this section, we propose a generic construction of identity-based matchmaking (IB-ME) encryption from IBS and IBE, following the "sign-then-encrypt" style.

To construct an IB-ME scheme $\mathsf{IB\text{-}ME} = (\mathsf{Setup}, \mathsf{SKeyGen}, \mathsf{RKeyGen}, \mathsf{Enc}, \mathsf{Dec})$ with the identity space $\mathcal{ID}$, message space $\mathcal{M}$ and ciphertext space $\mathcal{CT}$, we use the following two building blocks.

- An IBS scheme $\mathsf{IBS} = (\mathsf{IBS.Setup}, \mathsf{IBS.KeyGen}, \mathsf{IBS.Sign}, \mathsf{IBS.Ver})$ with the identity space $\mathcal{ID}$, message space $\mathcal{ID} \times \mathcal{ID} \times \mathcal{M}$ and signature space $\mathcal{S}$.
- An IBE scheme $\mathsf{IBE} = (\mathsf{IBE.Setup}, \mathsf{IBE.KeyGen}, \mathsf{IBE.Enc}, \mathsf{IBE.Dec})$ with the identity space $\mathcal{ID}$, message space $\mathcal{S} \times \mathcal{M}$ and ciphertext space $\mathcal{CT}$.

Our generic construction of IB-ME scheme IB-ME is described in Fig. 10. The correctness of IB-ME follows directly from the correctness of IBE and IBS.

Next we prove the ciphertext pseudo-randomness and the new stronger authenticity of our IB-ME. We show that the resulting IB-ME scheme has ciphertext pseudo-randomness if the underlying IBE has ciphertext pseudo-randomness, and it has the new stronger authenticity if the underlying IBS has EUF-ID-CMA security in the following theorem.

Theorem 5. *Let* IBE *be an IBE scheme satisfying ciphertext pseudo-randomness and* IBS *be an IBS scheme satisfying EUF-ID-CMA security. The scheme* IB-ME *given in Fig. 10 is an IB-ME scheme satisfying ciphertext pseudo-randomness and new stronger authenticity. More precisely, for any PPT adversaries $\mathcal{A}_1$ and $\mathcal{A}_2$, there exist PPT algorithms $\mathcal{B}_1$, $\mathcal{B}_2$ s.t.*

$$\mathsf{Adv}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}_1}(\kappa) \le \mathsf{Adv}^{cpse}_{\mathsf{IBE},\mathcal{B}_1}(\kappa), \quad \mathsf{Adv}^{new\text{-}saut}_{\mathsf{IB\text{-}ME},\mathcal{A}_2}(\kappa) \le \mathsf{Adv}^{EUF\text{-}ID\text{-}CMA}_{\mathsf{IBS},\mathcal{B}_2}(\kappa),$$

and $\mathcal{T}(\mathcal{B}_1) \approx \mathcal{T}(\mathcal{A}_1)$ and $\mathcal{T}(\mathcal{B}_2) \approx \mathcal{T}(\mathcal{A}_2)$.

Proof of ciphertext pseudo-randomness. Consider the ciphertext pseudo-randomness experiment $\mathsf{Exp}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}_1}(\kappa)$ defined in Fig. 7. We show that if there exists a PPT adversary $\mathcal{A}_1$ that wins in $\mathsf{Exp}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}_1}(\kappa)$, then there exists a PPT algorithm $\mathcal{B}_1$ that breaks the ciphertext pseudo-randomness of IBE.

Algorithm $\mathcal{B}_1$. Algorithm $\mathcal{B}_1$ is given $\mathsf{IBE}.mpk$ from its own challenger and has oracle access to $\mathcal{O}_{\mathsf{KeyGen}}$ as defined in Fig. 2. $\mathcal{B}_1$ simulates $\mathsf{Exp}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}_1}(\kappa)$ for $\mathcal{A}_1$ as follows.

0. During the initialization phase, the algorithm $\mathcal{B}_1$ initializes $\mathcal{Q}_{\mathsf{RKeyGen}} := \emptyset$.
1. During the setup phase, $\mathcal{B}_1$ invokes $(\mathsf{IBS}.mvk, \mathsf{IBS}.msk) \leftarrow \mathsf{IBS.Setup}(1^\kappa)$, sets $mpk := (\mathsf{IBE}.mpk, \mathsf{IBS}.mvk)$ and $msk := (\perp, \mathsf{IBS}.msk)$, and returns mpk to $\mathcal{A}_1$.

2. During the pre-challenge query phase, $\mathcal{B}_1$ proceeds as follows.
   - On receiving an SKeyGen query σ, $\mathcal{B}_1$ computes $\mathsf{IBS}.sk_\sigma \leftarrow \mathsf{IBS.KeyGen}(\mathsf{IBS}.msk, \sigma)$ and returns $ek_\sigma := (\mathsf{IBS}.sk_\sigma, \sigma)$.
   - On receiving an RKeyGen query ρ, $\mathcal{B}_1$ adds ρ to $\mathcal{Q}_{\mathsf{RKeyGen}}$, sends ρ as its query to $\mathcal{O}_{\mathsf{KeyGen}}$ and receives $\mathsf{IBE}.sk_\rho$. $\mathcal{B}_1$ returns $dk_\rho := (\mathsf{IBE}.sk_\rho, \rho)$.
3. On receiving the challenge query (σ^*, ρ^*, m^*) from $\mathcal{A}_1$, $\mathcal{B}_1$ invokes $\mathsf{IBS}.sk_{\sigma^*} \leftarrow \mathsf{IBS.KeyGen}(\mathsf{IBS}.msk, \sigma^*)$ to generate $ek_{\sigma^*} := (\mathsf{IBS}.sk_{\sigma^*}, \sigma^*)$, computes $\Sigma^* \leftarrow \mathsf{IBS.Sign}(\mathsf{IBS}.sk_{\sigma^*}, (\sigma^*, \rho^*, m^*))$ and sets $M^* := (\Sigma^*, m^*)$. $\mathcal{B}_1$ sends (ρ^*, M^*) as its own challenge and receives c^*. Finally, $\mathcal{B}_1$ returns c^* and ek_{σ^*} to $\mathcal{A}_1$.
4. During the post-challenge query phase, $\mathcal{B}_1$ proceeds as in step 2.
5. On receiving a guessing bit b' from $\mathcal{A}_1$, $\mathcal{B}_1$ checks if $\rho^* \in \mathcal{Q}_{\mathsf{RKeyGen}}$. If it is the case, $\mathcal{B}_1$ sends a random $b' \xleftarrow{\$} \{0,1\}$ to its challenger; otherwise, $\mathcal{B}_1$ sends b' to its challenger as its own guess.

Note that $\mathcal{B}_1$ implicitly shares the same challenge bit b with its challenger. If $b = 1$, c^* will be generated honestly by $\mathsf{IBE.Enc}(\mathsf{IBE}.mpk, \rho^*, M^*)$, then $\mathcal{B}_1$ will return the real generated ciphertext to $\mathcal{A}_1$. If $b = 0$, c^* will be sampled uniformly at random, then $\mathcal{B}_1$ will return a uniform ciphertext to $\mathcal{A}_1$. Hence $\mathcal{B}_1$ perfectly simulates $\mathsf{Exp}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}_1}(\kappa)$ for $\mathcal{A}_1$ and $\mathcal{B}_1$ wins if $\mathcal{A}_1$ wins. More precisely, we have:

$$\begin{aligned}\mathsf{Adv}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}_1}(\kappa) &= |\Pr[\mathsf{Exp}^{cpse}_{\mathsf{IB\text{-}ME},\mathcal{A}_1}(\kappa) \Rightarrow 1] - 1/2| \\ &\le |\Pr[\mathsf{Exp}^{cpse}_{\mathsf{IBE},\mathcal{B}_1}(\kappa) \Rightarrow 1] - 1/2| \le \mathsf{Adv}^{cpse}_{\mathsf{IBE},\mathcal{B}_1}(\kappa),\end{aligned}$$

and $\mathcal{T}(\mathcal{B}_1) \approx \mathcal{T}(\mathcal{A}_1)$. □

Proof of new stronger authenticity. Consider the new stronger authenticity experiment $\mathsf{Exp}^{new\text{-}saut}_{\mathsf{IB\text{-}ME},\mathcal{A}_2}(\kappa)$ defined in Fig. 8. We show that if there exists a PPT adversary $\mathcal{A}_2$ that wins in $\mathsf{Exp}^{new\text{-}saut}_{\mathsf{IB\text{-}ME},\mathcal{A}_2}(\kappa)$, then there exists a PPT algorithm $\mathcal{B}_2$ that breaks the EUF-ID-CMA security of IBS.

Algorithm $\mathcal{B}_2$**.** Algorithm $\mathcal{B}_2$ is given $\mathsf{IBS}.mvk$ from its own challenger and has oracle access to $\mathcal{O}_{\mathsf{KeyGen}}$ and $\mathcal{O}_{\mathsf{Sign}}$ as defined in Fig. 3. $\mathcal{B}_2$ simulates $\mathsf{Exp}^{new\text{-}saut}_{\mathsf{IB\text{-}ME},\mathcal{A}_2}(\kappa)$ for $\mathcal{A}_2$ as follows.

0. During the initialization phase, algorithm $\mathcal{B}_2$ initializes $\mathcal{Q}_{\mathsf{SKeyGen}} := \emptyset$, $\mathcal{Q}_{\mathsf{Enc}} := \emptyset$ and $\mathcal{Q}_{sk} := \emptyset$.
1. During the setup phase, $\mathcal{B}_2$ invokes $(\mathsf{IBE}.mpk, \mathsf{IBE}.msk) \leftarrow \mathsf{IBE.Setup}(1^\kappa)$, sets $mpk := (\mathsf{IBE}.mpk, \mathsf{IBS}.mvk)$ and $msk := (\mathsf{IBE}.msk, \perp)$, and returns mpk to $\mathcal{A}_2$.
2. During the query phase, $\mathcal{B}_2$ proceeds as follows.
   - On receiving an SKeyGen query σ, $\mathcal{B}_2$ adds σ to $\mathcal{Q}_{\mathsf{SKeyGen}}$ and checks if $\exists(ek_\sigma, \sigma) \in \mathcal{Q}_{sk}$. If yes, $\mathcal{B}_2$ returns ek_σ to $\mathcal{A}_2$; otherwise, $\mathcal{B}_2$ sends σ as its query to $\mathcal{O}_{\mathsf{KeyGen}}$ and receives $\mathsf{IBS}.sk_\sigma$, returns $ek_\sigma := (\mathsf{IBS}.sk_\sigma, \sigma)$ to $\mathcal{A}_2$ and adds (ek_σ, σ) to $\mathcal{Q}_{sk}$.
   - On receiving an RKeyGen query ρ, $\mathcal{B}_2$ invokes $\mathsf{IBE}.sk_\rho \leftarrow \mathsf{IBE.KeyGen}(\mathsf{IBE}.msk, \rho)$ and returns $dk_\rho := (\mathsf{IBE}.sk_\rho, \rho)$.
   - On receiving an Enc query (σ, ρ, m), $\mathcal{B}_2$ adds (σ, ρ, m) to $\mathcal{Q}_{\mathsf{Enc}}$, sends $(\sigma, (\sigma, \rho, m))$ as its query to $\mathcal{O}_{\mathsf{Sign}}$ and receives Σ. $\mathcal{B}_2$ returns $c \leftarrow \mathsf{IBE.Enc}(\mathsf{IBE}.mpk, \rho, (\Sigma, m))$ to $\mathcal{A}_2$.
3. On receiving the challenge query (c^*, ρ^*, σ^*) from $\mathcal{A}_2$, $\mathcal{B}_2$ generates $\mathsf{IBE}.sk_{\rho^*} \leftarrow \mathsf{IBE.KeyGen}(\mathsf{IBE}.msk, \rho^*)$ and computes $m' \leftarrow \mathsf{IBE.Dec}(\mathsf{IBE}.mpk, \mathsf{IBE}.sk_{\rho^*}, c^*)$. Parse $m' = (\Sigma^*, m^*)$. If $\sigma^* \in \mathcal{Q}_{\mathsf{SKeyGen}}$ or $(\sigma^*, \rho^*, m^*) \in \mathcal{Q}_{\mathsf{Enc}}$, $\mathcal{B}_2$ returns 0 to $\mathcal{A}_2$; otherwise, $\mathcal{B}_2$ returns $(\sigma^*, (\sigma^*, \rho^*, m^*), \Sigma^*)$ to its challenger.

Note that $\mathcal{B}_2$ perfectly simulates game $\mathsf{Exp}^{new\text{-}saut}_{\mathsf{IB\text{-}ME},\mathcal{A}_2}(\kappa)$ for $\mathcal{A}_2$. If $\mathcal{A}_2$ wins, it must hold that $m' = (\Sigma^*, m^*) \neq \perp$, $\sigma^* \notin \mathcal{Q}_{\mathsf{SKeyGen}}$, $(\sigma^*, \rho^*, m^*) \notin \mathcal{Q}_{\mathsf{Enc}}$ and $\mathsf{IBS.Ver}(\mathsf{IBS}.mvk, \sigma^*, (\sigma^*, \rho^*, m^*), \Sigma^*) = 1$. Hence σ^* is never submitted to $\mathcal{O}_{\mathsf{KeyGen}}$ and $(\sigma^*, (\sigma^*, \rho^*, m^*))$ is never submitted to $\mathcal{O}_{\mathsf{Sign}}$ by $\mathcal{B}_2$, and $(\sigma^*, (\sigma^*, \rho^*, m^*), \Sigma^*)$ passes the verification of IBS. Therefore, $(\sigma^*, (\sigma^*, \rho^*, m^*), \Sigma^*)$ is a valid forgery, and $\mathcal{B}_2$ wins in the EUF-ID-CMA experiment of IBS whenever $\mathcal{A}_2$ wins. So

$$\mathsf{Adv}^{new\text{-}saut}_{\mathsf{IB\text{-}ME},\mathcal{A}_2}(\kappa) = \Pr[\mathsf{Exp}^{new\text{-}saut}_{\mathsf{IB\text{-}ME},\mathcal{A}_2}(\kappa) \Rightarrow 1] \le \mathsf{Adv}^{\mathrm{EUF\text{-}ID\text{-}CMA}}_{\mathsf{IBS},\mathcal{B}_2}(\kappa),$$

and $\mathcal{T}(\mathcal{B}_2) \approx \mathcal{T}(\mathcal{A}_2)$. □

Remark 3. We note that the works [14,17] considered privacy against the target receiver, which is called "*enhanced privacy*" of IB-ME. In the definition of "*enhanced privacy*", the adversary can be the target receiver. However, it also has the following constraint:

- If $\mathcal{A}$ obtains the decryption key of target receiver rcv_0 (resp. rcv_1), then the distribution of sender set ID_0 (resp. ID_1) must have enough entropy.

In [17], the authors proposed using an extractor Ext to mask the ciphertext, thereby strengthening privacy to enhanced privacy. We remark that we can use a similar method for enhanced privacy. Concretely, a sender σ computes the ciphertext as $c \oplus \mathsf{Ext}(\sigma)$. Then, for any PPT adversary $\mathcal{A}$, even if $\mathcal{A}$ is the target receiver, it cannot distinguish $c \oplus \mathsf{Ext}(\sigma)$ from a uniform element when $\sigma \xleftarrow{\$} \mathsf{ID}$, since ID has enough entropy and thus $\mathsf{Ext}(\sigma)$ is statistically close to uniform, which masks c.

4.4. Instantiation of IB-ME from lattices

In this subsection, we instantiate our generic IB-ME construction using lattice-based IBE and IBS schemes. Specifically, we employ the LWE-based IBE scheme $\mathsf{IBE}_{\mathsf{BL}}$, proposed by Boyen and Li [20] and our proposed SIS-based IBS scheme in Section 3.3.

According to [20], we have the following Lemma.

Lemma 20 ([20]). *Let q, n, m be integer parameters s.t., $n, m = \mathsf{poly}(\kappa)$ and $q = O(\kappa^c) \cdot \omega(\sqrt{\log m})^2$ for some constant c. Let χ be a Gaussian distribution with parameter $\theta = \sqrt{n}$. Assuming that the $\mathsf{LWE}_{n,q,\chi,m}$ assumption holds and the existence of a secure pseudo-random function in NC^1, the $\mathsf{IBE}_{\mathsf{BL}}$ in [20] is a secure IBE scheme that satisfies the ciphertext pseudo-randomness.*[2] *More precisely, we have,*

$$\mathsf{Adv}^{cpse}_{\mathsf{IBE}_{\mathsf{BL}}}(\kappa) \leq 2\mathsf{Adv}^{\mathsf{LWE}}_{[n,q,\chi,m]}(\kappa) + 2\mathsf{Adv}^{pse}_{\mathsf{PRF}}(\kappa) + \mathsf{negl}(\kappa).$$

In this way, the resulting IB-ME is the IB-ME scheme instantiated by $\mathsf{IBE}_{\mathsf{BL}}$ and IBS in Fig. 5. By Theorems 5, 3, Lemmas 20 and 9, we have the following Corollary.

Corollary 1. *Let* IBS *be the IBS scheme given in Fig. 5 and* $\mathsf{IBE}_{\mathsf{BL}}$ *be the IBE scheme in [20]. Using* IBS *and* $\mathsf{IBE}_{\mathsf{BL}}$ *as building blocks, we construct an IB-ME scheme* IB-ME *that achieves ciphertext pseudo-randomness and new stronger authenticity.*

CRediT authorship contribution statement

Qing Yu: Writing – original draft, Methodology, Investigation, Formal analysis; **Shengli Liu:** Writing – review & editing, Supervision; **Yiming Li:** Writing – original draft, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

Appendix A. Proof of Theorem 1

Theorem 1. *The standard pseudo-randomness and the pseudo-randomness defined in Fig. 1 tightly imply each other. More specifically, for any PPT adversary $\mathcal{A}_1$ against the pseudo-randomness and adversary $\mathcal{A}_2$ against the standard pseudo-randomness (abbr. std-pse), there exist PPT adversaries $\mathcal{B}_1$ and $\mathcal{B}_2$ s.t.*

$$\mathsf{Adv}^{pse}_{\mathsf{PRF},\mathcal{A}_1}(\kappa) \leq 2 \cdot \mathsf{Adv}^{std\text{-}pse}_{\mathsf{PRF},\mathcal{B}_1}(\kappa),\ \mathsf{Adv}^{std\text{-}pse}_{\mathsf{PRF},\mathcal{A}_2}(\kappa) \leq \mathsf{Adv}^{pse}_{\mathsf{PRF},\mathcal{B}_2}(\kappa),$$

where $\mathcal{T}(\mathcal{B}_1) \approx \mathcal{T}(\mathcal{A}_1)$ and $\mathcal{T}(\mathcal{B}_2) \approx \mathcal{T}(\mathcal{A}_2)$.

Proof of Theorem 1. Clearly, if there is a PPT adversary $\mathcal{A}_2$ that breaks the standard pseudo-randomness, one can build a PPT algorithm $\mathcal{B}_2$ that breaks the pseudo-randomness: $\mathcal{B}_2$ simply forwards each query from $\mathcal{A}_2$ to its own challenger and relays the challenger's responses back to $\mathcal{A}_2$. Finally, $\mathcal{B}_2$ returns $\mathcal{A}_2$'s guess to the challenger. Hence, it suffices to focus on the first inequality.

To prove this, we define hybrid games G_0 and G_1, where G_0 is identical to $\mathsf{Exp}^{pse}_{\mathsf{PRF},\mathcal{A}_1}(\kappa)$ defined in Fig. 1. We show that G_0 and G_1 are indistinguishable. The differences between them are highlighted with gray shading.

Game G_0**.** Game G_0 is identical to $\mathsf{Exp}^{pse}_{\mathsf{PRF},\mathcal{A}_1}(\kappa)$ defined in Fig. 1.

0. During the initialization phase, the challenger $\mathcal{C}$ initializes sets $\mathcal{Q}_{\mathsf{PRF}} = \emptyset, \mathcal{Q}_{\mathsf{Chl}} = \emptyset$ and samples a bit $b \xleftarrow{\$} \{0,1\}$.
1. During the setup phase, $\mathcal{C}$ invokes $pp \leftarrow \mathsf{Setup}(1^\kappa)$, samples $k \xleftarrow{\$} \mathcal{K}$ and returns pp to $\mathcal{A}_1$.
2. During the pre-challenge query phase, $\mathcal{C}$ proceeds as follows.
   - On receiving a PRF query m, $\mathcal{C}$ adds m to $\mathcal{Q}_{\mathsf{PRF}}$ and returns $y \leftarrow \mathsf{PRF}(pp, k, m)$.
   - On receiving a Chl query m, $\mathcal{C}$ adds m to $\mathcal{Q}_{\mathsf{Chl}}$. If $b = 1$, $\mathcal{C}$ returns $y \leftarrow \mathsf{PRF}(pp, k, m)$; otherwise, $\mathcal{C}$ returns $y \leftarrow \mathsf{RF}(m)$.
3. On receiving the adversary's guess b' from $\mathcal{A}_1$, $\mathcal{C}$ checks if $\mathcal{Q}_{\mathsf{PRF}} \cap \mathcal{Q}_{\mathsf{Chl}} \neq \emptyset$. If so, G_0 outputs b. Otherwise, $\mathcal{C}$ compares b' to b: if $b = b'$, G_0 outputs 1; otherwise, G_0 outputs 0.

Game G_1**.** Game G_1 is similar to G_0 except that for any query from $\mathcal{A}_1$, the challenger $\mathcal{C}$ replies using the random function RF.

2′. During the pre-challenge query phase, $\mathcal{C}$ proceeds as follows.

[2] In [20] the authors prove that $\mathsf{IBE}_{\mathsf{BL}}$ achieves chosen-plaintext security (IND-ID-CPA); moreover, their proof implies that $\mathsf{IBE}_{\mathsf{BL}}$ satisfies ciphertext pseudo-randomness.

– On receiving a PRF query m, $\mathcal{C}$ adds m to $\mathcal{Q}_{\mathsf{PRF}}$ and returns $y \leftarrow \mathsf{RF}(m)$.
– On receiving a Chl query m, $\mathcal{C}$ adds m to $\mathcal{Q}_{\mathsf{Chl}}$ and returns $y \leftarrow \mathsf{RF}(m)$.

Lemma 21. $|\Pr[\mathsf{G}_0 \Rightarrow 1] - \Pr[\mathsf{G}_1 \Rightarrow 1]| \leq 2\mathsf{Adv}^{std\text{-}pse}_{\mathsf{PRF},\mathcal{B}_1}(\kappa)$.

Proof of Lemma 21. Suppose there exists a PPT adversary $\mathcal{A}_1$ that distinguishes games G_0 from G_1 with non-negligible advantage. We construct a PPT algorithm $\mathcal{B}_1$ that breaks the standard pseudo-randomness of the PRF.
Algorithm $\mathcal{B}_1$. Algorithm $\mathcal{B}_1$ is given the public parameter pp from its own challenger $\mathcal{C}$ and has oracle access to $\mathcal{O}_{\mathsf{Chl}}$ as defined in Fig. 1. $\mathcal{B}_1$ will simulate either G_0 or G_1 for $\mathcal{A}_1$ as follows.

0. During the initialization phase, the algorithm $\mathcal{B}_1$ initializes $\mathcal{Q}_{\mathsf{PRF}} = \emptyset, \mathcal{Q}_{\mathsf{Chl}} = \emptyset$ and samples a random bit $\beta \xleftarrow{\$} \{0,1\}$.
1. During the setup phase, $\mathcal{B}_1$ forwards pp to $\mathcal{A}_1$.
2. During the query phase, $\mathcal{B}_1$ proceeds as follows.
   – On receiving a PRF query m, $\mathcal{B}_1$ adds m to $\mathcal{Q}_{\mathsf{PRF}}$, forwards m to $\mathcal{O}_{\mathsf{Chl}}$, receives y and returns y to $\mathcal{A}_1$.
   – On receiving a Chl query m, $\mathcal{B}_1$ adds m to $\mathcal{Q}_{\mathsf{Chl}}$. If $m \in \mathcal{Q}_{\mathsf{PRF}}$, $\mathcal{B}_1$ forwards m to $\mathcal{O}_{\mathsf{Chl}}$, receives y and returns y to $\mathcal{A}_1$. Otherwise, if $\beta = 1$, $\mathcal{B}_1$ sends m as its query to $\mathcal{O}_{\mathsf{Chl}}$ and receives y; otherwise, sets $y \leftarrow \mathsf{RF}(m)$. Then $\mathcal{B}_1$ returns y to $\mathcal{A}_1$.
3. On receiving the adversary's guess β' from $\mathcal{A}_1$, $\mathcal{B}_1$ first checks if $\mathcal{Q}_{\mathsf{PRF}} \cap \mathcal{Q}_{\mathsf{Chl}} \neq \emptyset$. If so, $\mathcal{B}_1$ aborts and outputs β to its challenger. Otherwise, it compares β' to β: if $\beta' = \beta$, $\mathcal{B}_1$ returns $b' = 1$ to the challenger; otherwise, $\mathcal{B}_1$ returns $b' = 0$.

Without loss of generality, we assume that $\mathcal{A}_1$ never queries the same input to both oracles (i.e., $\mathcal{Q}_{\mathsf{PRF}} \cap \mathcal{Q}_{\mathsf{Chl}} = \emptyset$ always holds), since such an adversary has no advantage in distinguishing G_0 from G_1. By construction, if the challenger's bit $b = 1$, $\mathcal{B}_1$ perfectly simulates G_0 for $\mathcal{A}_1$; if $b = 0$, it simulates G_1. Therefore,

$$\begin{aligned}|\Pr[\mathsf{G}_0 \Rightarrow 1] - \Pr[\mathsf{G}_1 \Rightarrow 1]| &= |\Pr[\beta = \beta' | b = 1] - \Pr[\beta = \beta' | b = 0]| \\ &= |\Pr[b' = 1 | b = 1] - \Pr[b' = 1 | b = 0]| \\ &\leq 2\mathsf{Adv}^{std\text{-}pse}_{\mathsf{PRF},\mathcal{B}_1}(\kappa)\end{aligned}$$

□

Lemma 22. $\Pr[\mathsf{G}_1 \Rightarrow 1] = \frac{1}{2}$.

Proof of Lemma 22. In G_1, the challenger's responses are independent of b, so $\mathcal{A}_1$'s guess is correct with probability $\frac{1}{2}$. □

Combining Lemmas 21 and 22, we get

$$|\Pr[\mathsf{G}_0 \Rightarrow 1] - \frac{1}{2}| \leq 2\mathsf{Adv}^{std\text{-}pse}_{\mathsf{PRF},\mathcal{B}_1}(\kappa),$$

and $\mathcal{T}(\mathcal{B}_1) \approx \mathcal{T}(\mathcal{A}_1)$. □

Appendix B. Definition of Privacy for IB-ME

Privacy [2,15,16]. An IB-ME scheme IB-ME satisfies privacy if for all valid PPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, its advantage satisfies:

$$|\Pr[\mathsf{Exp}^{pri}_{\mathsf{IB\text{-}ME},\mathcal{A}}(\kappa) \Rightarrow 1] - 1/2| \leq \mathsf{negl}(\kappa),$$

where the security experiment $\mathsf{Exp}^{pri}_{\mathsf{IB\text{-}ME},\mathcal{A}}$ is defined in Fig. 11. Adversary $\mathcal{A}$ is called valid if $\forall \rho \in \mathcal{Q}_{\mathsf{RKeyGen}}$ it satisfies the following invariant:

(**Mismatch condition**) : $\rho \neq \mathsf{rcv}_0 \wedge \rho \neq \mathsf{rcv}_1$

$\underline{\mathsf{Exp}^{pri}_{\mathsf{IB\text{-}ME},\mathcal{A}}(\kappa)}$:
$(mpk, msk) \leftarrow \mathsf{Setup}(1^\kappa)$, $\mathcal{Q}_{\mathsf{RKeyGen}} := \emptyset$, $b \xleftarrow{\$} \{0,1\}$
$(\sigma_0, \sigma_1, \mathsf{rcv}_0, \mathsf{rcv}_1, m_0, m_1, St) \leftarrow \mathcal{A}_1^{\mathcal{O}_{\mathsf{SKeyGen}}(\cdot), \mathcal{O}_{\mathsf{RKeyGen}}(\cdot)}(mpk)$
$ek_{\sigma_b} \leftarrow \mathsf{SKeyGen}(mpk, msk, \sigma_b)$
$c \leftarrow \mathsf{Enc}(mpk, ek_{\sigma_b}, \mathsf{rcv}_b, m_b)$
$b' \leftarrow \mathcal{A}_2^{\mathcal{O}_{\mathsf{SKeyGen}}(\cdot), \mathcal{O}_{\mathsf{RKeyGen}}(\cdot)}(c, St)$
If $b = b'$: Return 1; Else: Return 0

$\underline{\mathcal{O}_{\mathsf{SKeyGen}}(\sigma)}$:
Return $ek_\sigma \leftarrow \mathsf{SKeyGen}(mpk, msk, \sigma)$

$\underline{\mathcal{O}_{\mathsf{RKeyGen}}(\rho)}$:
$\mathcal{Q}_{\mathsf{RKeyGen}} := \mathcal{Q}_{\mathsf{RKeyGen}} \cup \{\rho\}$
Return $dk_\rho \leftarrow \mathsf{RKeyGen}(mpk, msk, \rho)$

Fig. 11. Experiment $\mathsf{Exp}^{pri}_{\mathsf{IB\text{-}ME},\mathcal{A}}$ for defining the privacy of IB-ME.

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